

Sri Lanka Institute of Information Technology

Faculty of Computing

IT3021 — Data Warehousing and Business Intelligence

Assignment 01 — Full Report

New York State Hospital Inpatient Discharge Data Warehouse

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Task 1: Dataset Selection

1.1 Dataset Overview

The dataset selected is the New York State Inpatient Hospital Discharge Dataset (2009), sourced from the New York State Department of Health SPARCS (Statewide Planning and Research Cooperative System). The working dataset contains 49,999 patient admission records across 4 hospitals in the Albany area of New York State, with 34 attributes per record covering hospital information, patient demographics, clinical diagnoses and procedures, financial charges, and payment sources.

This is a genuine OLTP (Online Transaction Processing) dataset each row represents one real patient hospital admission transaction. It is not a pre-built analytics dataset and has not been used in any module practical sessions.

Key Dataset Statistics:

Attribute	Value
Total Records	49,999 patient admission transactions
Total Columns	34 attributes per record
Number of Hospitals	4 hospitals (Albany Medical Center, Albany Medical Center – South, Albany Memorial Hospital, St Peters Hospital)
Geographic Scope	Albany area, New York State - Health Service Area: Capital/Adiron
Time Period	Discharge Year: 2009 (all records)
Unique Diagnoses	253 CCS codes, 306 APR DRG codes, across 25 Major Diagnostic Categories
Unique Procedures	210 CCS procedure codes
Payment Types	10 payer types (Medicare, Medicaid, Blue Cross, Insurance, CHAMPUS, Workers Compensation, Self-Pay, etc.)
Total Charges Range	\$1,013.24 – \$1,451,544.10 (average: \$30,330.36)
Total Costs Range	\$181.44 – \$450,187.54 (average: \$10,685.61)
Length of Stay Range	1 – 113 days (average: 5.4 days)
Gender Values	M, F, U (Unknown - all three present in real data)
NULL Handling Needed	Source of Payment 3: 68.4% NULL; Source of Payment 2: 18.8% NULL; Zip Code: 13 NULLs

1.2 Why This Dataset is Suitable

- Multiple Source Types Required: Hospital info, clinical reference data, and transactions can be cleanly separated into a SQL Server database, an Excel workbook, and CSV files
- Rich Dimension Design: Supports 6 dimension tables (DimHospital, DimPatient, DimDiagnosis, DimProcedure, DimPayment, DimDate) all drawn from real attributes
- Natural Hierarchy: Diagnosis has a 3-level hierarchy — MDC (25 categories) → DRG (306 codes) → CCS Diagnosis (253 codes) — ideal for SSAS drill-down
- Slowly Changing Dimension: DimHospital is designed as SCD Type 2 — hospital names and service areas can change over time in a real healthcare system
- Measurable Fact Metrics: Three numeric measures (Length of Stay, Total Charges, Total Costs) are well-suited for aggregation, averaging, and trend analysis
- Realistic Data Quality Issues: NULLs in payment columns, Unknown gender values, and missing zip codes make this a real ETL challenge requiring transformation logic
- Sufficient Volume: 49,999 rows provide enough data for meaningful aggregations, SSAS cubes, and performance testing

1.3 All 34 Column Descriptions

The table below describes every column in the source dataset and maps it to its destination in the Data Warehouse:

Column Name	Description	DW Destination
Health Service Area	Geographic region the hospital belongs to	DimHospital
Hospital County	County where the hospital is located	DimHospital
Operating Certificate Number	NY State hospital operating license number	DimHospital
Facility ID	Unique numeric ID for the hospital facility	DimHospital
Facility Name	Full name of the hospital	DimHospital
Age Group	Patient age bracket (0-17, 18-29, 30-49, 50-69, 70+)	DimPatient
Zip Code - 3 digits	First 3 digits of patient's zip code	DimPatient
Gender	Patient gender: M, F, or U (Unknown)	DimPatient
Race	Patient race (White, Black/African American, Other)	DimPatient
Ethnicity	Spanish/Hispanic or Not Span/Hispanic	DimPatient

Length of Stay	Days admitted — range 1-113 days (Key Measure)	FactAdmission
Type of Admission	Emergency / Elective / Urgent / Newborn / Not Available	FactAdmission
Patient Disposition	How patient left (Home, Transfer, Rehab, etc.)	FactAdmission
Discharge Year	Year of discharge — only 2009 in this dataset	DimDate
CCS Diagnosis Code	Clinical Classification System diagnosis code (253 unique)	DimDiagnosis
CCS Diagnosis Description	Text description of the CCS diagnosis	DimDiagnosis
CCS Procedure Code	CCS procedure code (210 unique)	DimProcedure
CCS Procedure Description	Text description of the CCS procedure	DimProcedure
APR DRG Code	All Patient Refined Diagnosis Related Group (306 unique)	DimDiagnosis
APR DRG Description	Description of the APR DRG	DimDiagnosis
APR MDC Code	Major Diagnostic Category code (25 unique)	DimDiagnosis
APR MDC Description	Description of the Major Diagnostic Category	DimDiagnosis
APR Severity of Illness Code	Numeric severity: 1=Minor, 2=Moderate, 3=Major, 4=Extreme	DimDiagnosis
APR Severity of Illness Desc	Text: Minor / Moderate / Major / Extreme	DimDiagnosis
APR Risk of Mortality	Minor / Moderate / Major / Extreme	DimDiagnosis
APR Medical Surgical Description	Medical or Surgical classification	DimDiagnosis
Source of Payment 1	Primary payer (10 types including Workers Compensation)	DimPayment
Source of Payment 2	Secondary payer — 18.8% of records have this	DimPayment
Source of Payment 3	Tertiary payer — 68.4% NULL (most have none)	DimPayment
Birth Weight	Birth weight in grams; 0 if not a newborn admission	FactAdmission
Abortion Edit Indicator	Y/N — abortion edit flag	FactAdmission

Emergency Department Indicator	Y/N — whether Emergency Department was used	FactAdmission
Total Charges	Total charges billed (\$1,013 – \$1,451,544) — Key Measure	FactAdmission
Total Costs	Estimated hospital cost (\$181 – \$450,188) — Key Measure	FactAdmission

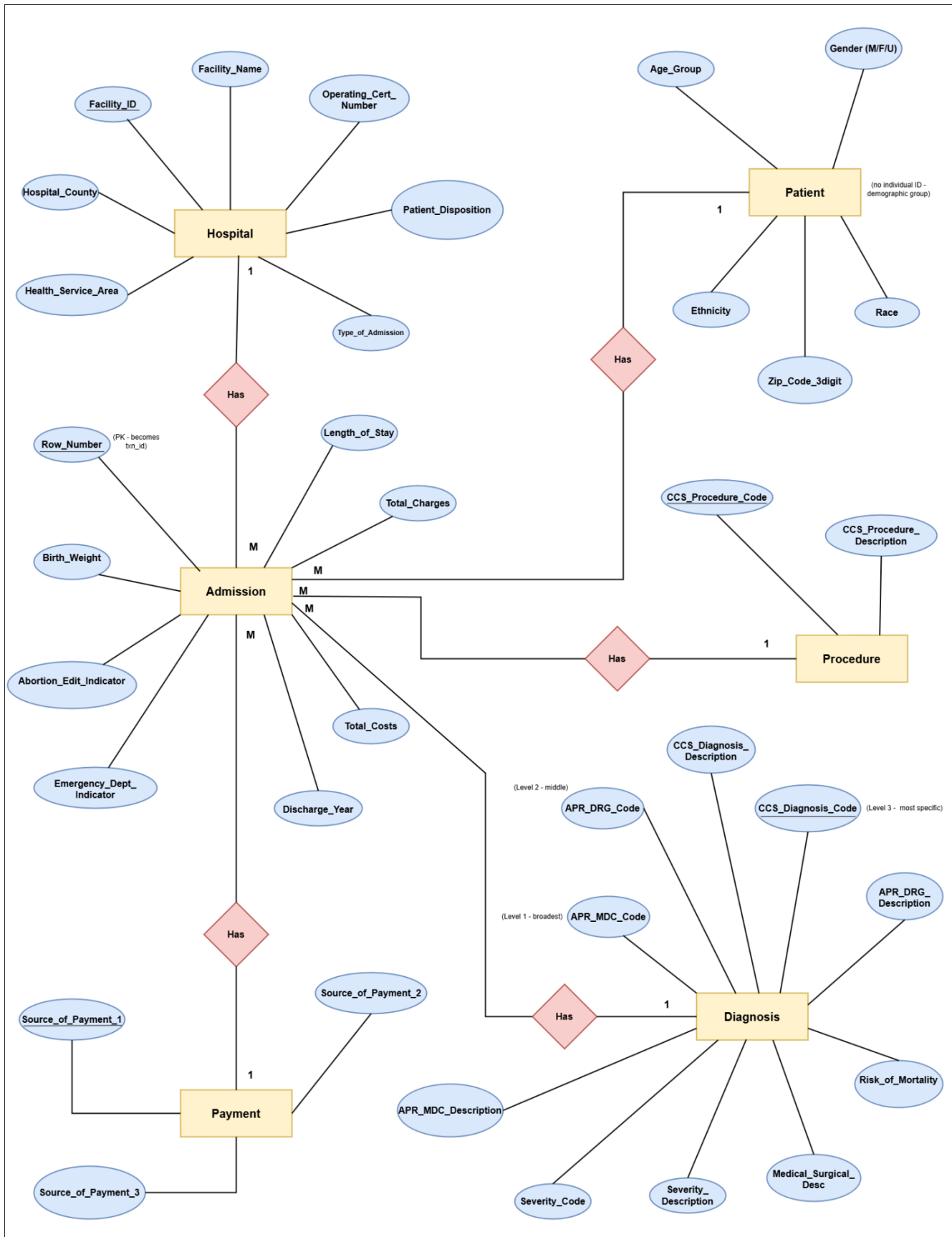
1.4 Entity-Relationship Diagram — Source Data Logical Entities

The source data is fully denormalized — all 34 columns exist in one flat row per admission. The logical entities and their relationships identified within this flat structure are:

Entity	Natural Key	Key Attributes
HOSPITAL	Facility ID	Facility Name, Operating Cert Number, Hospital County, Health Service Area (4 distinct hospitals)
PATIENT	Zip+Age+Gender+Race combo	Age Group, Gender (M/F/U), Race, Ethnicity, Zip Code (demographic grouping — no individual patient ID)
DIAGNOSIS	CCS Diagnosis Code	CCS Description, APR DRG Code/Desc, APR MDC Code/Desc, Severity (1–4), Risk of Mortality, Med/Surgical
PROCEDURE	CCS Procedure Code	CCS Procedure Description (210 unique procedures including 'NO PROC')
PAYMENT	Payment combination	Source of Payment 1 (always set), 2 (81% set), 3 (32% set) — 10 payer types
ADMISSION	(Row / txn_id)	Type of Admission, Patient Disposition, Length of Stay, Discharge Year, ED Indicator, Birth Weight, Abortion Flag, Total Charges, Total Costs

Relationships in source data: HOSPITAL has many ADMISSIONs. Each ADMISSION has one primary DIAGNOSIS, one PROCEDURE, and one PAYMENT record. PATIENT attributes describe the individual admitted. All relationships are represented as foreign keys in the Star Schema.

Entity-Relationship Diagram (ER):



1.5 Data Warehouse Star Schema Design

A Star Schema is used with FactAdmission at the centre. The grain is one row = one patient hospital admission/discharge event.

Fact Table: FactAdmission

FactAdmission — Grain: One row per patient admission		
Column Name	Data Type	Description
admission_key	INT IDENTITY (PK)	Surrogate primary key — auto-generated
hospital_key	INT (FK)	Foreign key → DimHospital
patient_key	INT (FK)	Foreign key → DimPatient
date_key	INT (FK)	Foreign key → DimDate
diagnosis_key	INT (FK)	Foreign key → DimDiagnosis
procedure_key	INT (FK)	Foreign key → DimProcedure
payment_key	INT (FK)	Foreign key → DimPayment
type_of_admission	VARCHAR(50)	Emergency / Elective / Urgent / Newborn / Not Available
patient_disposition	VARCHAR(100)	Patient discharge destination
emergency_dept_indicator	CHAR(1)	Y or N
abortion_edit_indicator	CHAR(1)	Y or N
birth_weight	INT	Birth weight in grams (0 if not newborn)
length_of_stay	INT	MEASURE: Days admitted (1 – 113)
total_charges	DECIMAL(12,2)	MEASURE: Total charges billed in USD
total_costs	DECIMAL(12,2)	MEASURE: Total cost to hospital in USD
accm_txn_create_time	DATETIME	Task 6: Timestamp when row was loaded into DW
accm_txn_complete_time	DATETIME NULL	Task 6: Timestamp when transaction was completed
txn_process_time_hours	DECIMAL(10,2) NULL	Task 6: Hours between create and complete

The three measures are: length_of_stay (additive — can be summed and averaged), total_charges, and total_costs. The three accm_ columns are added in Task 6 to implement an Accumulating Fact Table pattern.

DimHospital — Slowly Changing Dimension Type 2

DimHospital — SCD Type 2 (4 hospitals in dataset)		
Column Name	Data Type	Description
hospital_key	INT IDENTITY (PK)	Surrogate key
facility_id	INT	Natural key — original Facility ID (4 hospitals)
facility_name	VARCHAR(200)	Hospital name — SCD tracks name changes
operating_cert_number	VARCHAR(20)	NY State operating certificate number
hospital_county	VARCHAR(100)	County (Albany for all 4 hospitals)
health_service_area	VARCHAR(100)	Service region (Capital/Adiron for all)
effective_start_date	DATE	SCD Type 2: date this version became active
effective_end_date	DATE NULL	SCD Type 2: date this version was superseded
is_current	BIT	SCD Type 2: 1 = active record, 0 = historical

DimHospital is SCD Type 2. When a hospital's name or service area changes, a new row is inserted with a new effective_start_date, the old row's effective_end_date is updated, and is_current is set to 0 on the old row. This preserves the full history of hospital attribute changes. The natural key is facility_id; the surrogate key hospital_key is used in the fact table.

DimPatient

DimPatient — Demographic groupings (no individual patient ID in source)		
Column Name	Data Type	Description
patient_key	INT IDENTITY (PK)	Surrogate key
age_group	VARCHAR(20)	5 brackets: 0-17, 18-29, 30-49, 50-69, 70+
gender	CHAR(1)	M, F, or U (Unknown — found in full dataset)
race	VARCHAR(50)	White, Black/African American, Other Race
ethnicity	VARCHAR(50)	Spanish/Hispanic or Not Span/Hispanic
zip_code_3digit	VARCHAR(10)	3-digit zip code (13 NULLs → stored as 'Unknown')

No individual patient ID exists in the source data — demographics are modelled as a grouping dimension. Gender 'U' (Unknown) is included since the real full dataset contains this value.

DimDate

DimDate — Based on Discharge Year (2009 in this dataset)		
Column Name	Data Type	Description
date_key	INT (PK)	Surrogate key (e.g. 20090101)
discharge_year	INT	Year of discharge — only 2009 in this dataset
quarter	INT	Quarter (1–4) — derived
month	INT	Month (1–12) — derived
month_name	VARCHAR(20)	e.g. January — derived
year_quarter	VARCHAR(10)	e.g. 2009-Q1 — derived

DimDiagnosis — 3-level Hierarchy

DimDiagnosis — Hierarchy: MDC (25) → DRG (306) → CCS Diagnosis (253)		
Column Name	Data Type	Description
diagnosis_key	INT IDENTITY (PK)	Surrogate key
ccs_diagnosis_code	VARCHAR(10)	253 unique CCS diagnosis codes
ccs_diagnosis_description	VARCHAR(200)	CCS diagnosis text
apr_drg_code	VARCHAR(10)	306 unique APR DRG codes
apr_drg_description	VARCHAR(200)	APR DRG description
apr_mdc_code	VARCHAR(10)	25 Major Diagnostic Category codes
apr_mdc_description	VARCHAR(200)	MDC description — hierarchy level 2
severity_code	INT	1=Minor, 2=Moderate, 3=Major, 4=Extreme
severity_description	VARCHAR(20)	Minor / Moderate / Major / Extreme
risk_of_mortality	VARCHAR(20)	Minor / Moderate / Major / Extreme
medical_surgical_desc	VARCHAR(20)	Medical or Surgical

The 3-level hierarchy (MDC → DRG → CCS) enables SSAS drill-down: analyse at the body-system level (e.g. Cardiovascular), drill into DRG groups (e.g. Chest Pain), then drill to specific diagnoses. 1,918 unique MDC+DRG+CCS combinations exist in the full dataset.

DimProcedure

DimProcedure — 210 unique CCS procedure codes		
Column Name	Data Type	Description
procedure_key	INT IDENTITY (PK)	Surrogate key
ccs_procedure_code	VARCHAR(10)	210 unique CCS procedure codes
ccs_procedure_description	VARCHAR(200)	CCS procedure description

DimPayment

DimPayment — 10 payment source types + derived category		
Column Name	Data Type	Description
payment_key	INT IDENTITY (PK)	Surrogate key
source_of_payment_1	VARCHAR(100)	Primary payer — always populated
source_of_payment_2	VARCHAR(100) NULL	Secondary payer — 18.8% NULL
source_of_payment_3	VARCHAR(100) NULL	Tertiary payer — 68.4% NULL
payment_category	VARCHAR(50)	Grouped: Government / Private Insurance / Self-Pay / Unknown

DimPayment includes a derived payment_category column (Government / Private Insurance / Self-Pay / Unknown) that is not in the source data — it is added by joining to the PaymentCategoryMapping sheet in Source 3 during ETL.

1.6 Design Decisions and Assumptions

- No individual patient ID exists in the source data — patient demographics are modelled as a descriptive dimension (DimPatient) rather than a person entity
- Gender 'U' (Unknown) is present in the real full dataset and is handled explicitly — not treated as a NULL
- Type of Admission value 'Not Available' exists in the real data and is stored as-is in FactAdmission
- DimHospital is chosen as the SCD Type 2 dimension because hospital names and service areas change over time in a real healthcare system
- All three payment source columns are combined into one DimPayment dimension with an enrichment column (payment_category) sourced from the Excel lookup file
- Source of Payment 3 (68.4% NULL) and Source of Payment 2 (18.8% NULL) are stored as NULL in DimPayment — not replaced with 'Unknown' to preserve analytical accuracy
- The accumulating fact columns (accm_txn_create_time, accm_txn_complete_time, txn_process_time_hours) are included in the fact table schema from the start — they are populated in Task 6

Task 2: Preparation of Data Sources

2.1 Overview

The original dataset is a single flat Excel file (DWBI_Dataset.xlsx) containing all 34 columns in a denormalized format with 49,999 rows. To support realistic multi-source ETL development, the dataset has been split into four source files across three distinct file types: CSV, SQL Server Table, and Excel.

This separation mirrors a real healthcare IT environment: hospital administrative data lives in a relational database managed by IT, clinical coding reference tables are maintained in Excel by clinical teams, and transaction-level admission data arrives daily as CSV exports from the hospital's patient management system.

2.2 Source Files Summary

#	File Name	Type	Contents	Volume	DW Target
1	source1_admissions.csv	CSV File	Patient + admission transactions (all measures)	49,999 rows, 21 columns	FactAdmission, DimPatient, DimDate
2	source2_hospital_master.sql	SQL Server Table	Hospital reference — maintained in separate OLTP DB	4 hospitals, 7 columns	DimHospital (SCD Type 2)
3	source3_reference_data.xlsx	Excel Workbook	Diagnosis, procedure, payment lookups + MDC hierarchy	4 sheets, ~2,163 rows total	DimDiagnosis, DimProcedure, DimPayment
4	source4_txn_completions.csv	CSV File	Completion timestamps for Task 6 accumulating fact update	200 rows, 2 columns	FactAdmission (UPDATE)

2.3 Source 1 — Admissions Transactions (CSV)

File: [source1_admissions.csv](#) | 49,999 rows | 21 columns

This is the primary transaction source containing one row per patient hospital admission. It holds patient demographic attributes and all measurable admission data including the financial measures. The facility_id column links to the SQL Server hospital table (Source 2), and ccs_diagnosis_code and ccs_procedure_code link to the Excel reference workbook (Source 3). A txn_id is assigned to each row — this serves as the natural key for FactAdmission and is used by Source 4 in Task 6.

Column	Type	Notes
txn_id	INTEGER	Auto-generated row ID — natural key for fact table and Task 6 updates
facility_id	INTEGER	FK to HospitalMaster SQL table (4 hospitals in dataset)
discharge_year	INTEGER	Year of discharge — 2009 for all records in this dataset
age_group	VARCHAR	Patient age bracket (5 values)
zip_code_3digit	VARCHAR	3-digit zip code; NULL → replaced with 'Unknown'
gender	CHAR(1)	M / F / U (Unknown — present in full dataset)
race	VARCHAR	Patient race
ethnicity	VARCHAR	Hispanic status
length_of_stay	INTEGER	Days admitted (1–113 days range)
type_of_admission	VARCHAR	Emergency / Elective / Urgent / Newborn / Not Available
patient_disposition	VARCHAR	Patient discharge destination
ccs_diagnosis_code	VARCHAR	FK to DiagnosisReference sheet in Source 3 (253 unique codes)
ccs_procedure_code	VARCHAR	FK to ProcedureReference sheet in Source 3 (210 unique codes)
source_of_payment_1	VARCHAR	Primary payer — FK to PaymentCategoryMapping in Source 3
source_of_payment_2	VARCHAR	Secondary payer — empty string if NULL
source_of_payment_3	VARCHAR	Tertiary payer — 68.4% empty in real data
birth_weight	INTEGER	Birth weight (0 if not newborn)
abortion_edit_indicator	CHAR(1)	Y/N flag
emergency_dept_indicator	CHAR(1)	Y/N — was Emergency Dept used
total_charges	DECIMAL	Billed charges: \$1,013 – \$1,451,544 (avg \$30,330)
total_costs	DECIMAL	Hospital cost: \$181 – \$450,188 (avg \$10,686)

Data Quality Actions Applied in Source 1:

- zip_code_3digit: 13 NULL values replaced with 'Unknown'
- source_of_payment_2 and _3: NULL values stored as empty string (to be handled as NULL in SSIS)
- gender: 'U' (Unknown) values preserved — not treated as errors
- type_of_admission: 'Not Available' values preserved as-is

2.4 Source 2 — Hospital Master Data (SQL Server Table)

File: [source2_hospital_master.sql](#) | 4 hospitals | Run in SSMS → creates NYHospitalOLTP database

Hospital master reference data is stored in a SQL Server OLTP database (NYHospitalOLTP). This represents a hospital registration system maintained separately by the IT department. The 4 real hospitals from the dataset are included: Albany Medical Center Hospital, Albany Medical Center – South Clinical Campus, Albany Memorial Hospital, and St Peters Hospital — all in Albany County, Health Service Area Capital/Adiron.

This source feeds DimHospital, which is implemented as an SCD Type 2 dimension. In SSIS, this source is accessed via an OLE DB Source connection to the NYHospitalOLTP database.

Column	Type	Description
facility_id	INT (PK)	Unique hospital ID — natural key (4 hospitals)
facility_name	VARCHAR	Full hospital name
operating_cert_number	VARCHAR	NY State operating certificate
hospital_county	VARCHAR	County (Albany for all 4 in this dataset)
health_service_area	VARCHAR	Geographic service region (Capital/Adiron for all)
record_created_date	DATE	When this record was added to the system
is_active	BIT	1 = active, 0 = decommissioned hospital

2.5 Source 3 — Reference & Lookup Data (Excel Workbook)

File: `source3_reference_data.xlsx` | 4 sheets | ~2,163 rows total | Derived from full 49,999-row dataset

Clinical coding reference data and enrichment lookup tables are maintained in an Excel workbook. This reflects how clinical coding teams typically manage reference tables in a real healthcare setting. All values in this workbook are extracted from or derived from the actual 49,999-row dataset — there are no invented values.

Sheet Name	Rows	Contents
DiagnosisReference	1,918 rows	253 CCS codes × DRG/MDC combinations: codes, descriptions, severity, risk, med/surg
ProcedureReference	210 rows	All 210 unique CCS procedure codes with descriptions
PaymentCategoryMapping	10 rows	10 payment sources mapped to Government / Private Insurance / Self-Pay / Unknown
MDCHierarchy	25 rows	25 MDC codes mapped to descriptions and body system groups — enables SSAS hierarchy

Why Excel for Source 3:

- DiagnosisReference has 1,918 rows covering all real CCS × DRG combinations found in the full 49,999-row dataset
- ProcedureReference has all 210 real procedure codes from the dataset
- PaymentCategoryMapping introduces an enrichment attribute (`payment_category`) not in the original data — adds analytical value
- MDCHierarchy enables the 3-level diagnosis hierarchy in SSAS: Body System Group → MDC → DRG → CCS Diagnosis
- In SSIS, this source is read using an Excel Source component via an Excel Connection Manager

2.6 Source 4 — Transaction Completions (CSV, Task 6)

File: [source4_txn_completions.csv](#) | 200 rows | Used exclusively in Task 6 Accumulating Fact Table

This file is used in Task 6 to implement the Accumulating Fact Table pattern. It simulates a downstream administrative system that sends completion timestamps once hospital admission transactions are fully processed (1–7 days after the admission date). The SSIS package in Task 6 reads this file, looks up the matching FactAdmission row by `txn_id`, and updates the `accm_txn_complete_time` and `txn_process_time_hours` columns.

Column	Type	Description
txn_id	INTEGER	Matches <code>txn_id</code> in Source 1 — used to UPDATE the correct FactAdmission row
accm_txn_complete_time	DATETIME	Timestamp of transaction completion (1–7 days after admission date)

2.7 Full Attribute-to-Source Mapping

Attribute(s)	Source File	Type	DW Target
txn_id, discharge_year, length_of_stay, total_charges, total_costs, type_of_admission, patient_disposition, ed_indicator, birth_weight, abortion_indicator	source1_admissions.csv	CSV	FactAdmission
age_group, gender, race, ethnicity, zip_code_3digit	source1_admissions.csv	CSV	DimPatient
facility_id, facility_name, operating_cert, hospital_county, health_service_area	source2_hospital_master.sql	SQL Server Table	DimHospital (SCD2)
ccs_diagnosis_code/desc, apr_drg_code/desc, apr_mdc_code/desc, severity, risk, med_surg	source3 → DiagnosisReference	Excel	DimDiagnosis

ccs_procedure_code, ccs_procedure_description	source3 → ProcedureReference	Excel	DimProcedure
source_of_payment_1/2/3 (values from Source 1) + payment_category (lookup from Source 3)	source1 + source3 → PaymentCategoryMapping	CSV + Excel	DimPayment
discharge_year + derived: quarter, month, month_name, year_quarter	source1_admissions.csv	CSV	DimDate
txn_id, accm_txn_complete_time	source4_txn_completions.csv	CSV	FactAdmission (UPDATE)

2.8 Summary

Four source files of three distinct types have been prepared using actual values from the full 49,999-row dataset — no data has been invented or approximated:

- CSV (Sources 1 and 4): Transaction data (49,999 rows) and Task 6 completion updates (200 rows)
- SQL Server Table (Source 2): 4 real hospitals in a relational database — accessed via OLE DB in SSIS
- Excel Workbook (Source 3): 4 sheets covering 1,918 diagnosis combinations, 210 procedures, 10 payment type mappings, and 25 MDC hierarchy entries

The three source types ensure the SSIS ETL pipeline demonstrates reading from flat files (Flat File Source), querying a relational database (OLE DB Source), and parsing Excel sheets (Excel Source). The enrichment columns added in Source 3 (payment_category, body_system_group) introduce analytical value beyond the original flat data.

Task 3: Solution Architecture

3.1 Overview

The DW/BI solution for the New York State Hospital Inpatient Discharge system follows a classic five-layer architecture: Data Sources → Staging Area → ETL (SSIS) → Data Warehouse → BI & Reporting. This architecture ensures clean separation of concerns — raw data is isolated in staging, transformation logic is centralised in SSIS packages, and the data warehouse serves only clean, structured analytical data to reporting tools.

All components are implemented on the Microsoft SQL Server stack: SQL Server Database Engine for the staging area and data warehouse, SQL Server Integration Services (SSIS) for ETL, SQL Server Analysis Services (SSAS) for OLAP cubes, and SQL Server Reporting Services (SSRS) for reports.

3.2 Architecture Diagram

The diagram below shows the five-layer architecture with all components, data flows, and the tools used at each layer:

DW/BI Solution Architecture — NY Hospital Inpatient Discharge System

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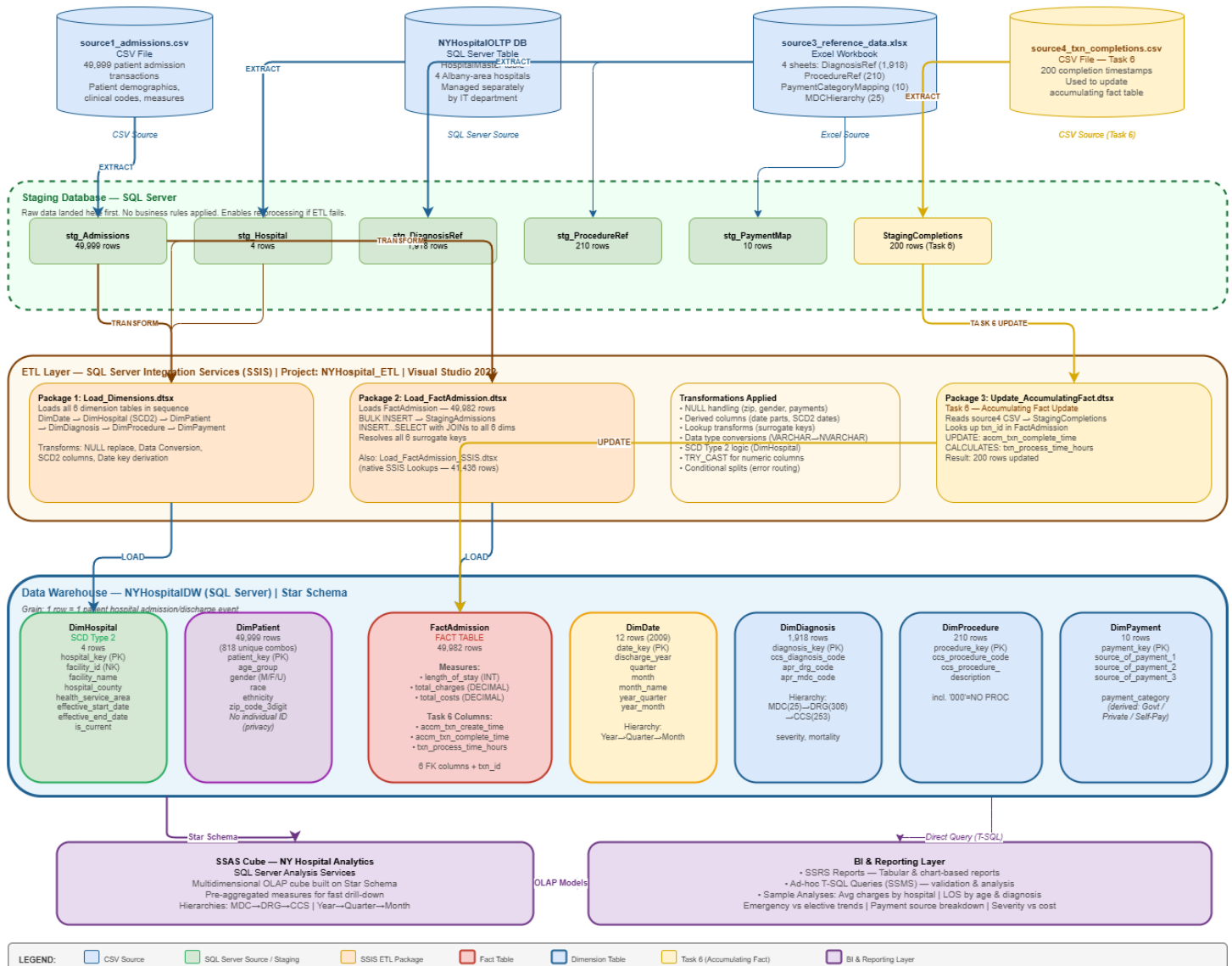


Figure 3.0: DW/BI Solution Architecture — NY Hospital Inpatient Discharge System

IT3021 – DATA WAREHOUSING & BUSINESS INTELLIGENCE

DW/BI Solution Architecture — NY Hospital Inpatient Discharge System



IT3021 Assignment 1 – SLIIT Year 3 Semester 1 – 2026 – NY Hospital Inpatient Discharge DW/BI System

Figure 3.1: DW/BI Solution Architecture — NY Hospital Inpatient Discharge System

3.3 Component Descriptions

The following sections describe each layer of the architecture in detail.

1. Data Sources

The origin points of all data fed into the DW/BI system. Contains three source types: (1) CSV files with 49,999 patient admission transactions and Task 6 completion updates, (2) a SQL Server OLTP database (NYHospitalOLTP) holding hospital master data for 4 Albany-area hospitals, and (3) an Excel workbook with 4 sheets of clinical reference data including diagnoses, procedures, payment mappings, and the MDC hierarchy. Using multiple source types demonstrates real-world ETL complexity.

2. Staging Area

An intermediate SQL Server database where raw data is landed exactly as extracted from sources — no transformations applied. Staging serves as a buffer: if the ETL fails partway through, data can be re-processed from staging without re-extracting from sources. The staging area holds tables mirroring each source: stg_Admissions (49,999 rows), stg_Hospital (4 rows), stg_DiagnosisRef (1,918 rows), stg_ProcedureRef (210 rows), and stg_PaymentMap (10 rows). NULL values are preserved in staging.

3. ETL Layer (SSIS)

The transformation engine built using SQL Server Integration Services (SSIS) in Visual Studio. Contains three SSIS packages: Package 1 loads all six dimension tables in the correct order (DimHospital with SCD Type 2 logic, DimPatient, DimDate, DimDiagnosis, DimProcedure, DimPayment). Package 2 loads FactAdmission by reading staging data, performing surrogate key lookups against all dimension tables, and writing the fact rows. Package 3 (Task 6) reads the completion CSV and updates FactAdmission accumulating columns. Transformations include NULL replacement, derived column calculations, data type conversions, lookup transforms, conditional splits for error routing, and SCD Type 2 change detection for DimHospital.

4. Data Warehouse

The central analytical database built on a Star Schema. FactAdmission sits at the centre with 49,982 rows and three measures (length_of_stay, total_charges, total_costs). Six dimension tables surround it: DimHospital (SCD Type 2 — preserves history of hospital name/region changes), DimPatient (demographic groupings), DimDate (Year→Quarter→Month hierarchy), DimDiagnosis (3-level hierarchy: MDC→DRG→CCS), DimProcedure (210 procedure types), and DimPayment (10 payer types with enriched payment category). All tables use integer surrogate keys. The fact table also includes three accumulating columns for Task 6.

5. BI & Reporting

The consumption layer where business users and analysts access the data. SQL Server Analysis Services (SSAS) builds a multidimensional OLAP cube on top of the Star Schema, pre-aggregating measures and enabling fast drill-down across the diagnosis hierarchy (MDC→DRG→CCS) and date hierarchy (Year→Quarter→Month). SQL Server Reporting Services (SSRS) generates tabular and chart-based reports such as average total charges by hospital, length of stay by age group, and admissions by payment type. Ad-hoc T-SQL queries in SSMS allow direct validation and custom analysis of the data warehouse.

3.4 Data Flow Description

Data moves through the architecture in the following sequence:

- **Step 1**
Extract: SSIS packages connect to all three source types simultaneously. The Flat File Source component reads source1_admissions.csv (49,999 rows). The OLE DB Source connects to NYHospitalOLTP and reads the HospitalMaster table (4 rows). The Excel Source reads all four sheets from source3_reference_data.xlsx.
- **Step 2**
Stage: All extracted data is written to staging tables in the staging database with no transformations applied. This creates a reliable recovery point and allows the ETL to be re-run if needed.
- **Step 3**
Transform: SSIS Package 1 reads from staging and applies all transformations to load dimension tables: NULL replacement for zip codes and gender, derived column calculations for date parts (quarter, month), SCD Type 2 logic for DimHospital (detecting changes and inserting new versions), lookup joins to enrich payment data with payment_category from the Excel mapping sheet.
- **Step 4**
Load Dimensions: All six dimension tables are loaded in the correct order. DimDate is populated first (no dependencies), then DimHospital, DimPatient, DimDiagnosis, DimProcedure, and DimPayment. Surrogate keys are auto-generated by SQL Server IDENTITY columns.
- **Step 5**
Load Fact: SSIS Package 2 reads stg_Admissions and performs lookup transforms against each dimension table to resolve surrogate keys (hospital_key, patient_key, date_key, diagnosis_key, procedure_key, payment_key). The accm_txn_create_time is set to GETDATE() at load time. Completed rows are written to FactAdmission.
- **Step 6**

Accumulating Update (Task 6): SSIS Package 3 reads source4_txn_completions.csv, looks up each txn_id in FactAdmission, and executes an OLE DB Command to UPDATE the accm_txn_complete_time. A Derived Column component then calculates txn_process_time_hours as the difference in hours between the two timestamps.

- **Step 7**

Analyse & Report: SSAS builds a cube on the Star Schema. SSRS consumes the cube and the DW directly to produce reports. Business users can drill down through the diagnosis hierarchy (MDC → DRG → CCS) and time hierarchy (Year → Quarter → Month).

3.5 Technology Stack

Component	Technology	Purpose
Data Sources	CSV, SQL Server, Excel	Raw OLTP data in three formats
Staging Database	SQL Server Database	Intermediate landing zone for raw extracted data
ETL Engine	SQL Server SSIS (Visual Studio)	Extract, transform, and load all data
Data Warehouse	SQL Server Database	Star Schema — FactAdmission + 6 dimension tables
OLAP Cube	SQL Server SSAS	Pre-aggregated multidimensional analysis with hierarchies
Reports	SQL Server SSRS	Tabular reports, charts, and dashboards
Query & Validation	SSMS / T-SQL	Ad-hoc queries, validation, and troubleshooting

3.6 Key Design Decisions

- **Staging Area Included:** Although not strictly required, a staging area is included to follow best practice. It enables safe re-processing if ETL fails and makes debugging easier by separating extraction from transformation.
- **Dimensions Before Facts:** SSIS packages are designed so all dimension tables are fully loaded before FactAdmission is loaded. This ensures all foreign key lookups succeed and no fact rows are orphaned.
- **SCD Type 2 on DimHospital:** DimHospital uses Slowly Changing Dimension Type 2 to preserve the full history of hospital attribute changes (name, county, service area). This is important for accurate historical reporting — an admission in 2009 must always refer to the hospital's 2009 attributes.

- Star Schema over Snowflake: A Star Schema is chosen for simplicity, query performance, and ease of SSAS cube design. The diagnosis hierarchy (MDC→DRG→CCS) is stored in a single DimDiagnosis table rather than being snowflaked into separate tables.
- Separate Package for Task 6: The accumulating fact table update is implemented as a separate SSIS package (Package 3) rather than modifying Package 2. This follows the principle of separation of concerns and allows the update to be run independently when completion data arrives.

Task 4: Data Warehouse Design & Development

4.1 Dimensional Model Overview

The data warehouse uses a Star Schema. FactAdmission sits at the centre surrounded by six dimension tables. The schema is implemented in a SQL Server database named NYHospitalDW.

Item	Value	Notes
Schema Type	Star Schema	Single fact table surrounded by denormalized dimensions
Fact Table	FactAdmission	49,982 rows — one per patient admission event
Dimension Tables	6 tables	DimHospital, DimPatient, DimDate, DimDiagnosis, DimProcedure, DimPayment
Grain	1 admission	One row = one patient hospital admission/discharge event
Measures	3 measures	length_of_stay, total_charges, total_costs
SCD	Type 2	DimHospital — preserves full history of hospital changes
Hierarchy	3-level	DimDiagnosis: MDC (25) → DRG (306) → CCS Diagnosis (253)
Database	NYHospitalDW	SQL Server — created by Task4_Create_DataWarehouse.sql

4.2 Star Schema Diagram

The diagram below shows the Star Schema with all tables, keys, and relationships. The fact table is at the centre and all dimension tables connect directly to it.

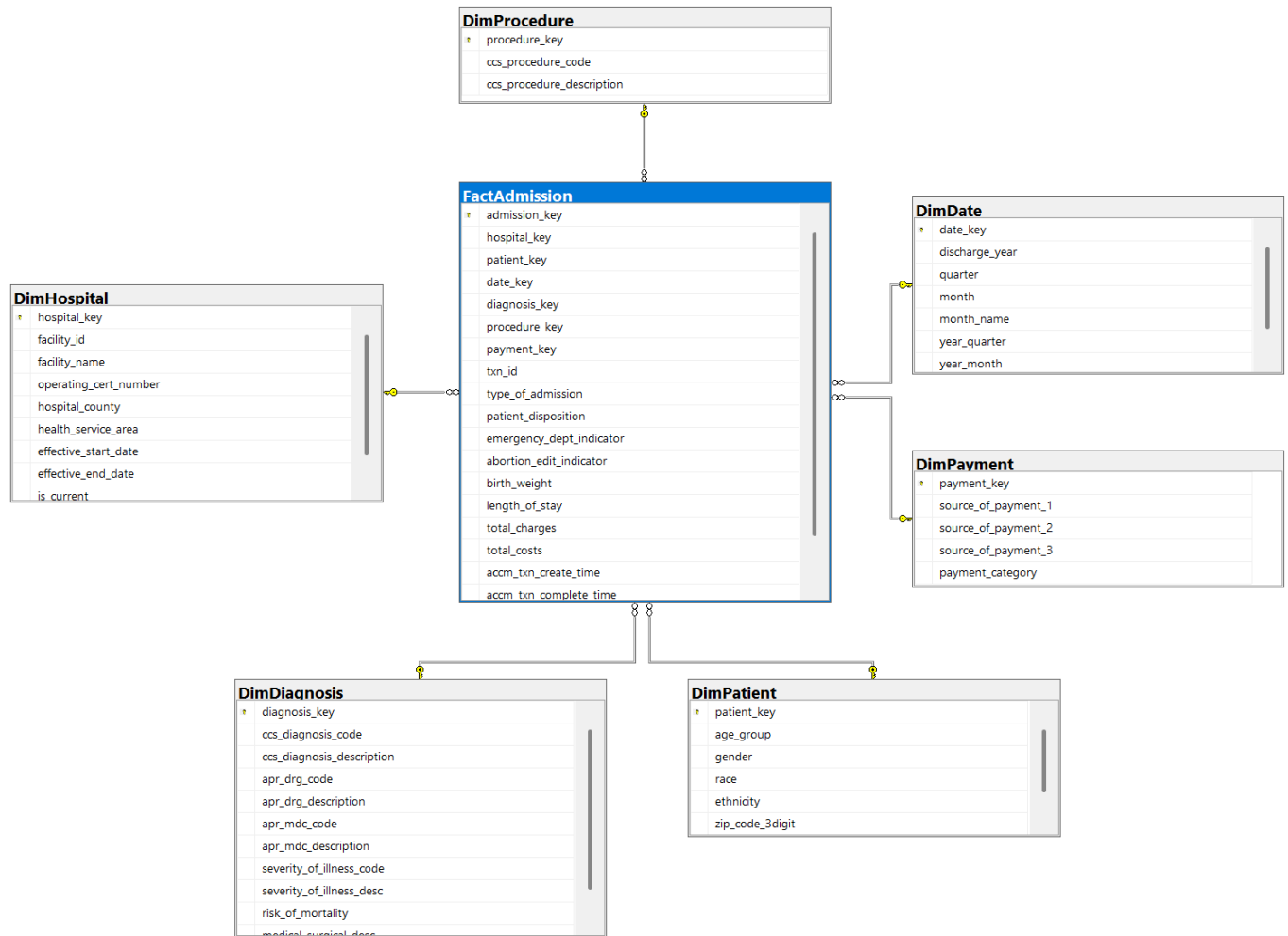
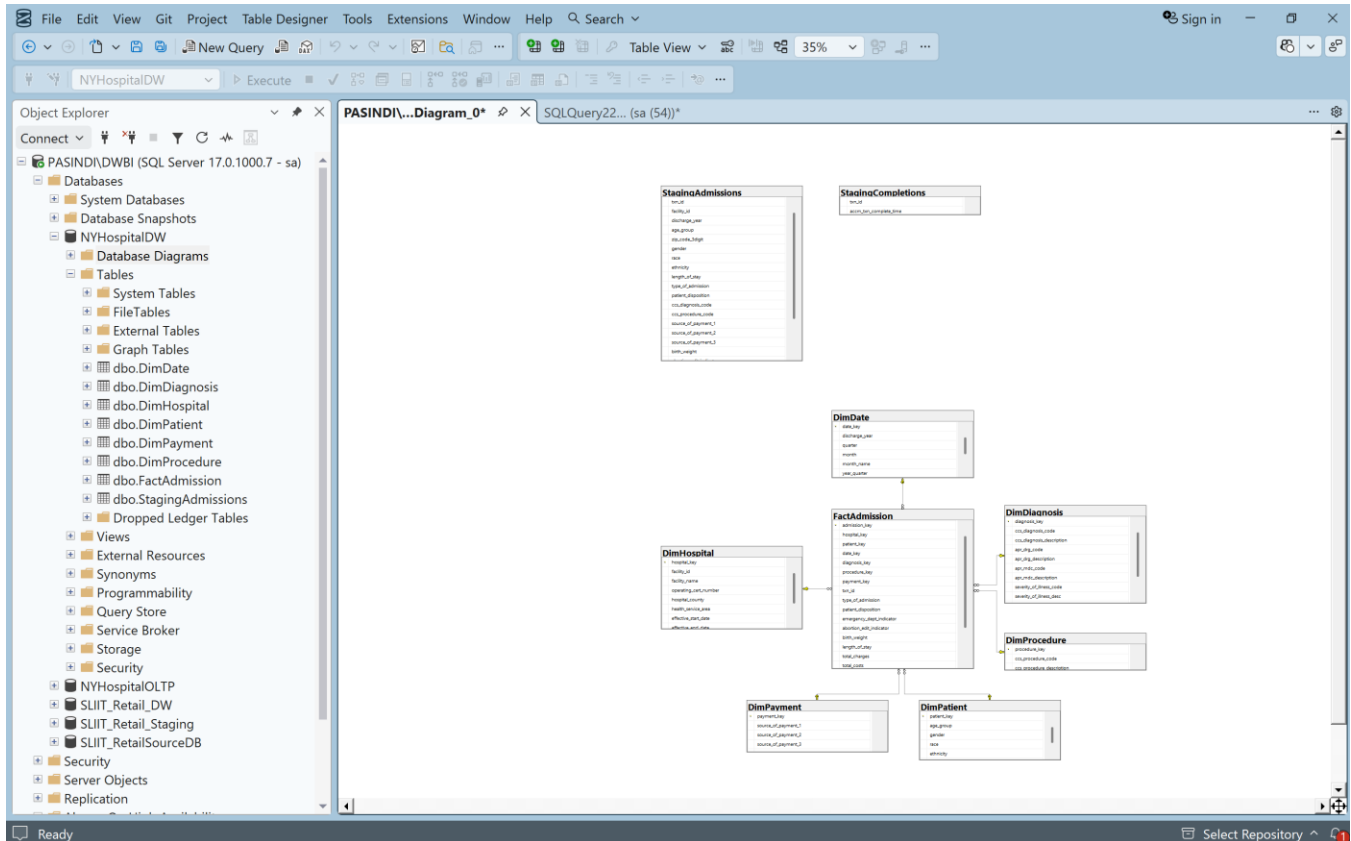


Figure 4.1: Star Schema — NYHospitalDW (NY Hospital Inpatient Discharge)

Snapshot of Database Daigram :



4.3 Table Definitions

4.3.1 FactAdmission — Fact Table

Grain: One row per patient hospital admission/discharge event. Contains 3 numeric measures and 6 foreign keys to dimension tables. Also contains Task 6 accumulating columns.

FactAdmission			
Column Name	Data Type	Key	Description
admission_key	INT	PK IDENTITY	Surrogate primary key — auto-generated
hospital_key	INT NOT NULL	FK	→ DimHospital.hospital_key
patient_key	INT NOT NULL	FK	→ DimPatient.patient_key
date_key	INT NOT NULL	FK	→ DimDate.date_key
diagnosis_key	INT NOT NULL	FK	→ DimDiagnosis.diagnosis_key
procedure_key	INT NOT NULL	FK	→ DimProcedure.procedure_key
payment_key	INT NOT NULL	FK	→ DimPayment.payment_key
txn_id	INT NOT NULL	Natural Key	Source row number — used for Task 6 updates
type_of_admission	VARCHAR(20)		Emergency / Elective / Urgent / Newborn / Not Available
patient_disposition	VARCHAR(50)		18 possible values e.g. 'Home or Self Care', 'Expired'
emergency_dept_indicator	CHAR(1)		Y or N
abortion_edit_indicator	CHAR(1)		Y or N
birth_weight	INT		Grams — 0 if not newborn (max 9,900g in dataset)
length_of_stay	INT	MEASURE	Days admitted — range 1 to 120 (avg 5.4)
total_charges	DECIMAL(12,2)	MEASURE	Total billed charges — \$1,013 to \$1,451,544 (avg \$30,330)
total_costs	DECIMAL(12,2)	MEASURE	Hospital cost — \$181 to \$450,188 (avg \$10,686)
accm_txn_create_time	DATETIME	Task 6	Set to GETDATE() when row is loaded by SSIS Package 2
accm_txn_complete_time	DATETIME NULL	Task 6	Updated by SSIS Package 3 when completion CSV is processed

txn_process_time_hours	DECIMAL(10,2)	Task 6	Calculated: (accm_txn_complete_time - accm_txn_create_time) in hours
------------------------	---------------	--------	--

4.3.2 DimHospital — Slowly Changing Dimension Type 2

Tracks hospital master data. When a hospital's name or service area changes, a new row is inserted with updated dates and the old row is marked as historical (is_current = 0). The natural key is facility_id; the surrogate key hospital_key is used in FactAdmission.

DimHospital (SCD Type 2)			
Column Name	Data Type	Key	Description
hospital_key	INT	PK IDENTITY	Surrogate key
facility_id	INT NOT NULL	Natural Key	Original hospital ID from source (4 hospitals: IDs 1,2,4,5)
facility_name	VARCHAR(100)		Hospital name — SCD2 tracks changes to this value
operating_cert_number	VARCHAR(20)		NY State operating certificate number
hospital_county	VARCHAR(50)		County (Albany for all 4 hospitals in dataset)
health_service_area	VARCHAR(50)		Region (Capital/Adiron for all hospitals in dataset)
effective_start_date	DATE	SCD2	Date this version of the record became active
effective_end_date	DATE NULL	SCD2	Date superseded — NULL means currently active
is_current	BIT	SCD2	1 = active current record; 0 = historical version

SCD Type 2 Logic: When SSIS detects a change in facility_name, hospital_county, or health_service_area for an existing facility_id, it: (1) UPDATES the old row — sets effective_end_date = today, is_current = 0. (2) INSERTs a new row with the new values, effective_start_date = today, effective_end_date = NULL, is_current = 1.

4.3.3 DimPatient

No individual patient ID exists in the source data. Patient demographics are stored as a grouping dimension — each unique combination of age group, gender, race, ethnicity, and zip code gets one row.

DimPatient			
Column Name	Data Type	Key	Description
patient_key	INT	PK IDENTITY	Surrogate key
age_group	VARCHAR(20)		5 values: '0 to 17', '18 to 29', '30 to 49', '50 to 69', '70 or Older'
gender	CHAR(1)		M, F, or U (Unknown — present in real dataset)
race	VARCHAR(50)		White, Black/African American, Other Race, Unknown
ethnicity	VARCHAR(30)		Not Span/Hispanic, Spanish/Hispanic, Unknown
zip_code_3digit	VARCHAR(10)		3-digit zip; 'Unknown' substituted for 13 NULL values in source

4.3.4 DimDate

Based on the Discharge Year column. Since all 49,999 records are from year 2009, DimDate is pre-populated with 12 rows (one per month of 2009). Quarter and month are derived to enable Year → Quarter → Month hierarchy in SSAS.

DimDate			
Column Name	Data Type	Key	Description
date_key	INT	PK	Surrogate key format YYYYMMDD01 (e.g. 20090101)
discharge_year	INT NOT NULL		Year of discharge — only 2009 in this dataset
quarter	INT NOT NULL		Quarter: 1, 2, 3, or 4

month	INT NOT NULL		Month: 1 to 12
month_name	VARCHAR(20)		January, February, ... December
year_quarter	VARCHAR(10)		e.g. '2009-Q1', '2009-Q2' — for SSAS hierarchy
year_month	VARCHAR(10)		e.g. '2009-01' — for monthly reporting

4.3.5 DimDiagnosis — 3-Level Hierarchy

Contains 1,918 unique combinations of CCS Diagnosis Code × APR DRG Code × APR MDC Code. The three levels form a hierarchy used for SSAS drill-down: MDC (25 categories) → DRG (306 codes) → CCS Diagnosis (253 codes).

DimDiagnosis (Hierarchy: MDC → DRG → CCS)			
Column Name	Data Type	Key	Description
diagnosis_key	INT	PK IDENTITY	Surrogate key
ccs_diagnosis_code	VARCHAR(10)		253 unique codes e.g. '083' (EPILEPSY/CONVULSIONS)
ccs_diagnosis_description	VARCHAR(50)		CCS diagnosis text — max 23 chars in dataset
apr_drg_code	VARCHAR(10)		306 unique APR DRG codes
apr_drg_description	VARCHAR(150)		APR DRG description — max 89 chars
apr_mdc_code	VARCHAR(10)		25 Major Diagnostic Category codes (top of hierarchy)
apr_mdc_description	VARCHAR(150)		MDC description — max 100 chars
severity_of_illness_code	INT		1=Minor, 2=Moderate, 3=Major, 4=Extreme
severity_of_illness_desc	VARCHAR(20)		Minor / Moderate / Major / Extreme
risk_of_mortality	VARCHAR(20)		Minor / Moderate / Major / Extreme
medical_surgical_desc	VARCHAR(20)		Medical or Surgical

4.3.6 DimProcedure

Contains 210 unique CCS procedure codes. Includes code '000' with description 'NO PROC' — representing admissions where no procedure was performed.

DimProcedure			
Column Name	Data Type	Key	Description
procedure_key	INT	PK IDENTITY	Surrogate key
ccs_procedure_code	VARCHAR(10)	Unique	210 unique codes incl. '000' = NO PROC
ccs_procedure_description	VARCHAR(50)		CCS procedure text — max 23 chars in dataset

4.3.7 DimPayment

Stores the three payment source columns from the source data plus a derived payment_category column. The payment_category value is looked up from the PaymentCategoryMapping sheet in Source 3 during ETL, grouping the 10 raw payer names into 4 categories.

DimPayment			
Column Name	Data Type	Key	Description
payment_key	INT	PK IDENTITY	Surrogate key
source_of_payment_1	VARCHAR(50)	NOT NULL	Primary payer — always populated (10 payer types)
source_of_payment_2	VARCHAR(50)	NULL	Secondary payer — NULL for 18.8% of admissions
source_of_payment_3	VARCHAR(50)	NULL	Tertiary payer — NULL for 68.4% of admissions
payment_category	VARCHAR(30)	Derived	Government / Private Insurance / Self-Pay / Unknown — from Source 3 lookup

4.4 Execution Order

Tables must be created in this order because the fact table has foreign key constraints to all dimension tables:

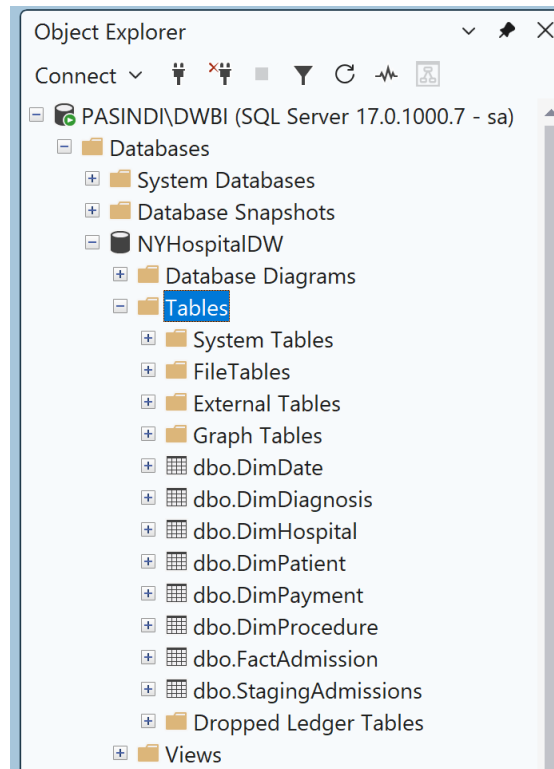
Step	Table	Reason
1	DimDate	No foreign key dependencies — must exist before FactAdmission
2	DimHospital	No foreign key dependencies — must exist before FactAdmission
3	DimPatient	No foreign key dependencies — must exist before FactAdmission
4	DimDiagnosis	No foreign key dependencies — must exist before FactAdmission
5	DimProcedure	No foreign key dependencies — must exist before FactAdmission
6	DimPayment	No foreign key dependencies — must exist before FactAdmission
7	FactAdmission	LAST — has FK constraints to all 6 dimension tables

4.5 Design Assumptions

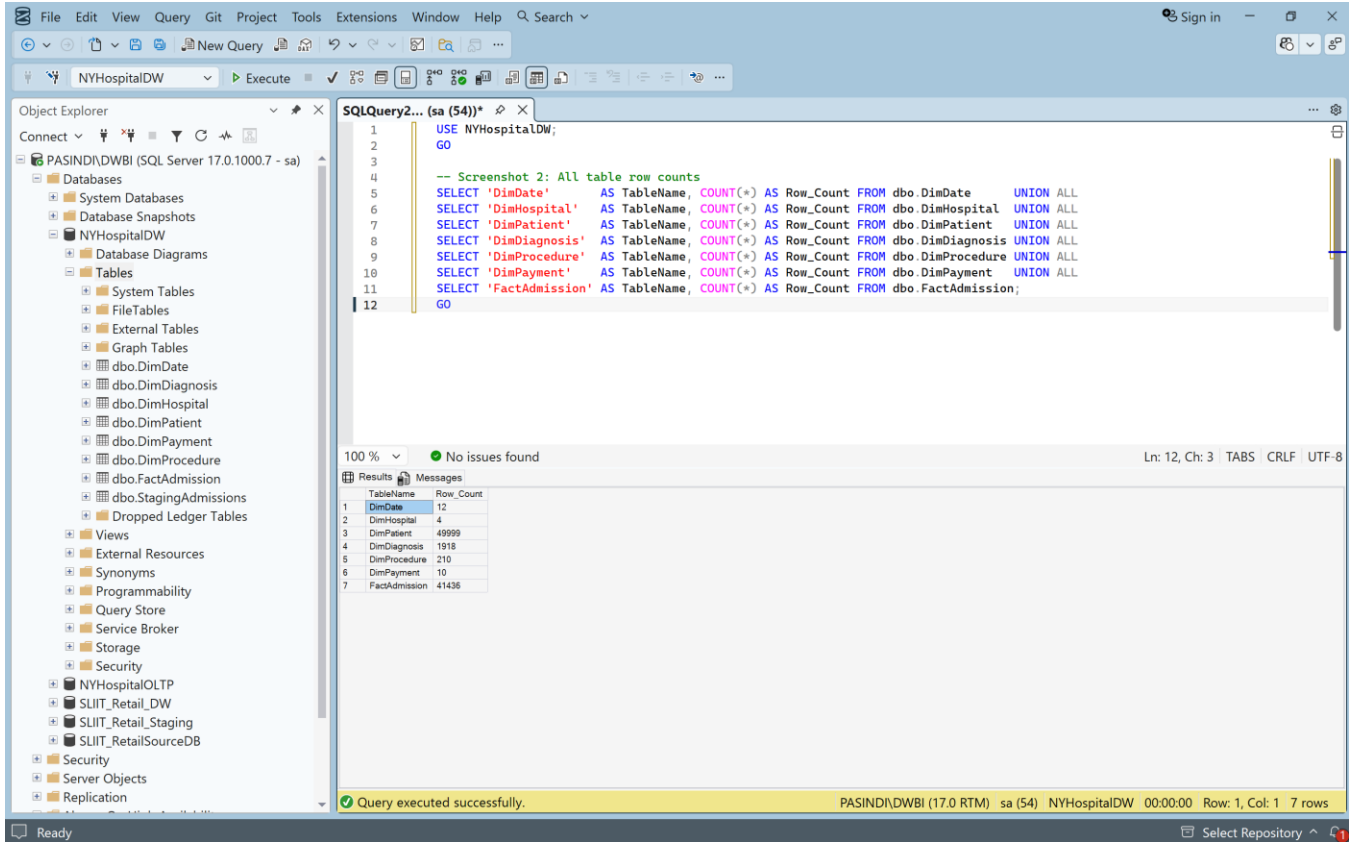
- No individual patient identifier exists in the source data. The patient dimension groups records by demographic attributes (age, gender, race, ethnicity, zip). This is standard practice in healthcare data where patient privacy must be protected.
- All 49,999 records are from discharge year 2009. DimDate is pre-populated with 12 monthly rows for 2009 rather than being dynamically generated. Quarter and month values are derived attributes, not stored in the source.
- DimHospital is the SCD Type 2 dimension because hospital names and service areas can change over time in a real healthcare system. All other dimensions are treated as SCD Type 1 (overwrite on change) for simplicity.
- The three payment source columns are combined into one DimPayment dimension rather than creating separate payment dimensions. The derived payment_category column adds analytical value not present in the source.
- type_of_admission and patient_disposition are stored as degenerate dimensions (directly in FactAdmission) because they describe the admission event itself and would not support useful standalone analysis as separate dimension tables.
- VARCHAR sizes are based on actual maximum observed values in the real 49,999-row dataset — not arbitrary sizes. For example patient_disposition is VARCHAR(50) because the longest value is 37 characters.
- The txn_id column in FactAdmission stores the natural key (row number from source1_admissions.csv) to enable the Task 6 accumulating fact table update package to locate specific rows for updating.

4.6 Verification Screenshots

Screenshot 1: Object Explorer showing all 7 tables created



Screenshot 2: Verification Query 1 — Table row counts



The screenshot shows the SQL Server Enterprise Manager interface. The Object Explorer on the left displays the database structure for PASINDI\DWBI (SQL Server 17.0.1000.7 - sa). The central query window, titled 'SQLQuery2... (sa (54))', contains the following SQL script:

```

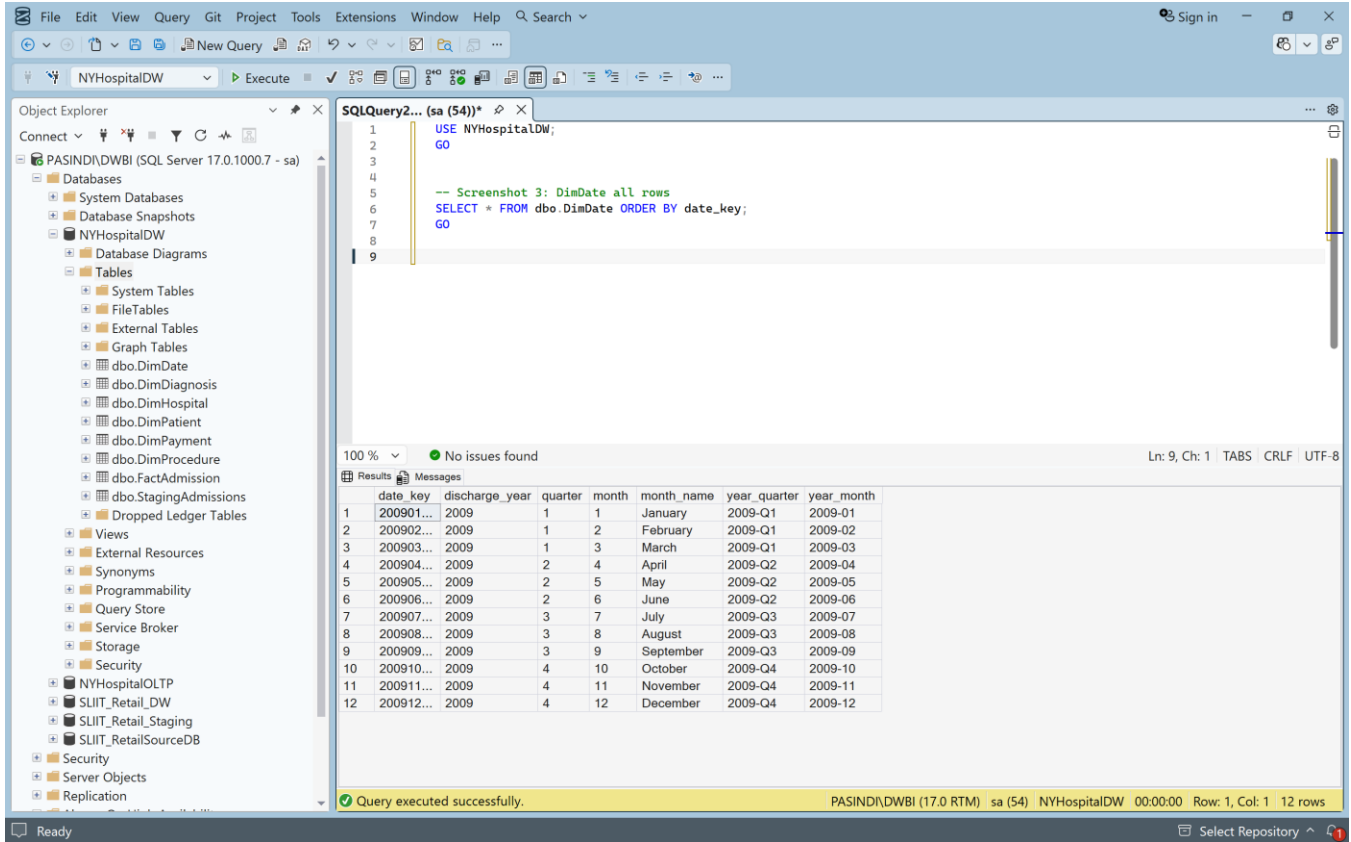
1  USE NYHospitalDW;
2  GO
3
4  -- Screenshot 2: All table row counts
5  SELECT 'DimDate' AS TableName, COUNT(*) AS Row_Count FROM dbo.DimDate UNION ALL
6  SELECT 'DimHospital' AS TableName, COUNT(*) AS Row_Count FROM dbo.DimHospital UNION ALL
7  SELECT 'DimPatient' AS TableName, COUNT(*) AS Row_Count FROM dbo.DimPatient UNION ALL
8  SELECT 'DimDiagnosis' AS TableName, COUNT(*) AS Row_Count FROM dbo.DimDiagnosis UNION ALL
9  SELECT 'DimProcedure' AS TableName, COUNT(*) AS Row_Count FROM dbo.DimProcedure UNION ALL
10 SELECT 'DimPayment' AS TableName, COUNT(*) AS Row_Count FROM dbo.DimPayment UNION ALL
11 SELECT 'FactAdmission' AS TableName, COUNT(*) AS Row_Count FROM dbo.FactAdmission;
12 GO
  
```

The Results pane at the bottom displays the output of the query, showing the row counts for each table:

TableName	Row_Count
DimDate	12
DimHospital	4
DimPatient	49999
DimDiagnosis	1918
DimProcedure	210
DimPayment	10
FactAdmission	41436

The status bar at the bottom indicates that the query was executed successfully. The message pane shows: "Query executed successfully. PASINDI\DWBI (17.0 RTM) sa (54) NYHospitalDW 00:00:00 Row: 1, Col: 1 7 rows".

Screenshot 3: DimDate populated



The screenshot shows the SQL Server Enterprise Manager interface. The Object Explorer on the left displays the database structure for PASINDI\DWBI (SQL Server 17.0.1000.7 - sa). The database NYHospitalDW is expanded, showing the dbo schema with various tables. The table dbo.DimDate is highlighted.

The SQL Query window (SQLQuery2... (sa (54))*) contains the following query:

```

1  USE NYHospitalDW;
2  GO
3
4
5  -- Screenshot 3: DimDate all rows
6  SELECT * FROM dbo.DimDate ORDER BY date_key;
7  GO
8
9

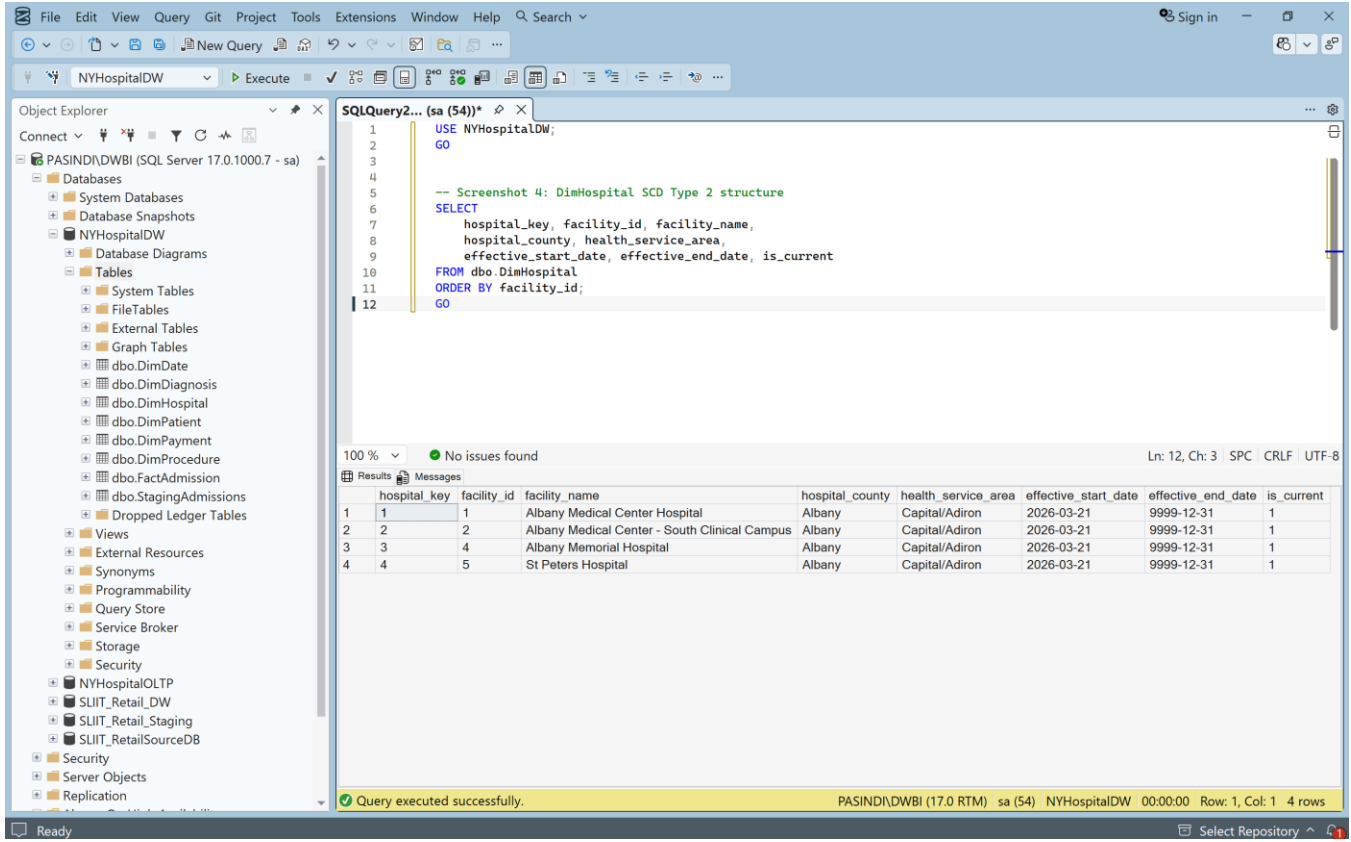
```

The Results pane shows the output of the query, displaying 12 rows of data. The columns are: date_key, discharge_year, quarter, month, month_name, year_quarter, and year_month.

	date_key	discharge_year	quarter	month	month_name	year_quarter	year_month
1	200901...	2009	1	1	January	2009-Q1	2009-01
2	200902...	2009	1	2	February	2009-Q1	2009-02
3	200903...	2009	1	3	March	2009-Q1	2009-03
4	200904...	2009	2	4	April	2009-Q2	2009-04
5	200905...	2009	2	5	May	2009-Q2	2009-05
6	200906...	2009	2	6	June	2009-Q2	2009-06
7	200907...	2009	3	7	July	2009-Q3	2009-07
8	200908...	2009	3	8	August	2009-Q3	2009-08
9	200909...	2009	3	9	September	2009-Q3	2009-09
10	200910...	2009	4	10	October	2009-Q4	2009-10
11	200911...	2009	4	11	November	2009-Q4	2009-11
12	200912...	2009	4	12	December	2009-Q4	2009-12

The status bar at the bottom indicates: Query executed successfully. PASINDI\DWBI (17.0 RTM) sa (54) NYHospitalDW 00:00:00 Row: 1, Col: 1 12 rows

Screenshot 4: DimHospital SCD Type 2 structure



The screenshot displays the SQL Server Enterprise Manager interface. The left pane shows the Object Explorer with the database structure of NYHospitalDW. The right pane shows a SQL query window with the following code:

```

1  USE NYHospitalDW;
2  GO
3
4
5  -- Screenshot 4: DimHospital SCD Type 2 structure
6
7  SELECT
8      hospital_key, facility_id, facility_name,
9      hospital_county, health_service_area,
10     effective_start_date, effective_end_date, is_current
11  FROM dbo.DimHospital
12  ORDER BY facility_id;
13  GO
  
```

Below the query window, the results of the query are displayed in a table. The table has 8 columns: hospital_key, facility_id, facility_name, hospital_county, health_service_area, effective_start_date, effective_end_date, and is_current. The results show 4 rows of data.

hospital_key	facility_id	facility_name	hospital_county	health_service_area	effective_start_date	effective_end_date	is_current
1	1	Albany Medical Center Hospital	Albany	Capital/Adiron	2026-03-21	9999-12-31	1
2	2	Albany Medical Center - South Clinical Campus	Albany	Capital/Adiron	2026-03-21	9999-12-31	1
3	4	Albany Memorial Hospital	Albany	Capital/Adiron	2026-03-21	9999-12-31	1
4	5	St Peters Hospital	Albany	Capital/Adiron	2026-03-21	9999-12-31	1

The status bar at the bottom indicates that the query was executed successfully, returning 4 rows.

Task 5: ETL Development using SSIS

5.1 Overview

The ETL process for the NYHospitalDW Data Warehouse was implemented using SQL Server Integration Services (SSIS) in Visual Studio 2022. The ETL consists of two main SSIS packages and supplementary SQL scripts that extract data from three source types, transform it, and load it into the Star Schema.

5.2 SSIS Project Setup

- Project Name: NYHospital_ETL
- Project Type: Integration Services Project (Visual Studio 2022)
- Location: E:\Year 3\Y3 S1\DWBI - IT3021\dataset\NYHospital_ETL
- Packages: Load_Dimensions.dtsx, Load_FactAdmission.dtsx

Connection Managers Created:

Connection Name	Type	Connects To
NYHospitalDW	OLE DB	NYHospitalDW database — Data Warehouse destination
NYHospitalOLTP	OLE DB	NYHospitalOLTP database — Hospital master source
Source1_Admissions	Flat File	source1_admissions.csv — 49,999 admission transactions
Source3_Reference	Excel	source3_reference_data.xlsx — 4 reference sheets

5.3 Package 1: Load_Dimensions.dtsx

Package 1 loads all six dimension tables in the correct order. Dimensions must be loaded before the fact table to ensure foreign key lookups succeed. The Control Flow consists of 6 Data Flow Tasks connected in sequence with precedence constraints.

Control Flow — Execution Order:

#	Task	Source	Key Transformations
1	Load DimDate	NYHospitalDW (DimDate already populated)	Reads existing DimDate — 12 rows pre-loaded by SQL script for year 2009
2	Load DimHospital	NYHospitalOLTP (HospitalMaster table)	OLE DB Source → Data Conversion (VARCHAR→NVARCHAR) → Derived Column (SCD2 dates) → OLE DB Destination. Adds effective_start_date, effective_end_date, is_current
3	Load DimPatient	Source1_Admissions.csv	Flat File Source → Data Conversion (to NVARCHAR) → OLE DB Destination. Loads 49,999 demographic rows
4	Load DimDiagnosis	Source3_Reference.xlsx (DiagnosisReference)	Excel Source → Data Conversion (code columns to NVARCHAR) → OLE DB Destination. Loads 1,918 diagnosis combinations
5	Load DimProcedure	Source3_Reference.xlsx (ProcedureReference)	Excel Source → OLE DB Destination. Loads 210 procedure codes
6	Load DimPayment	Source3_Reference.xlsx (PaymentCategoryMapping)	Excel Source → OLE DB Destination. Loads 10 payment types with categories

5.3.1 Screenshot — Package 1 Control Flow

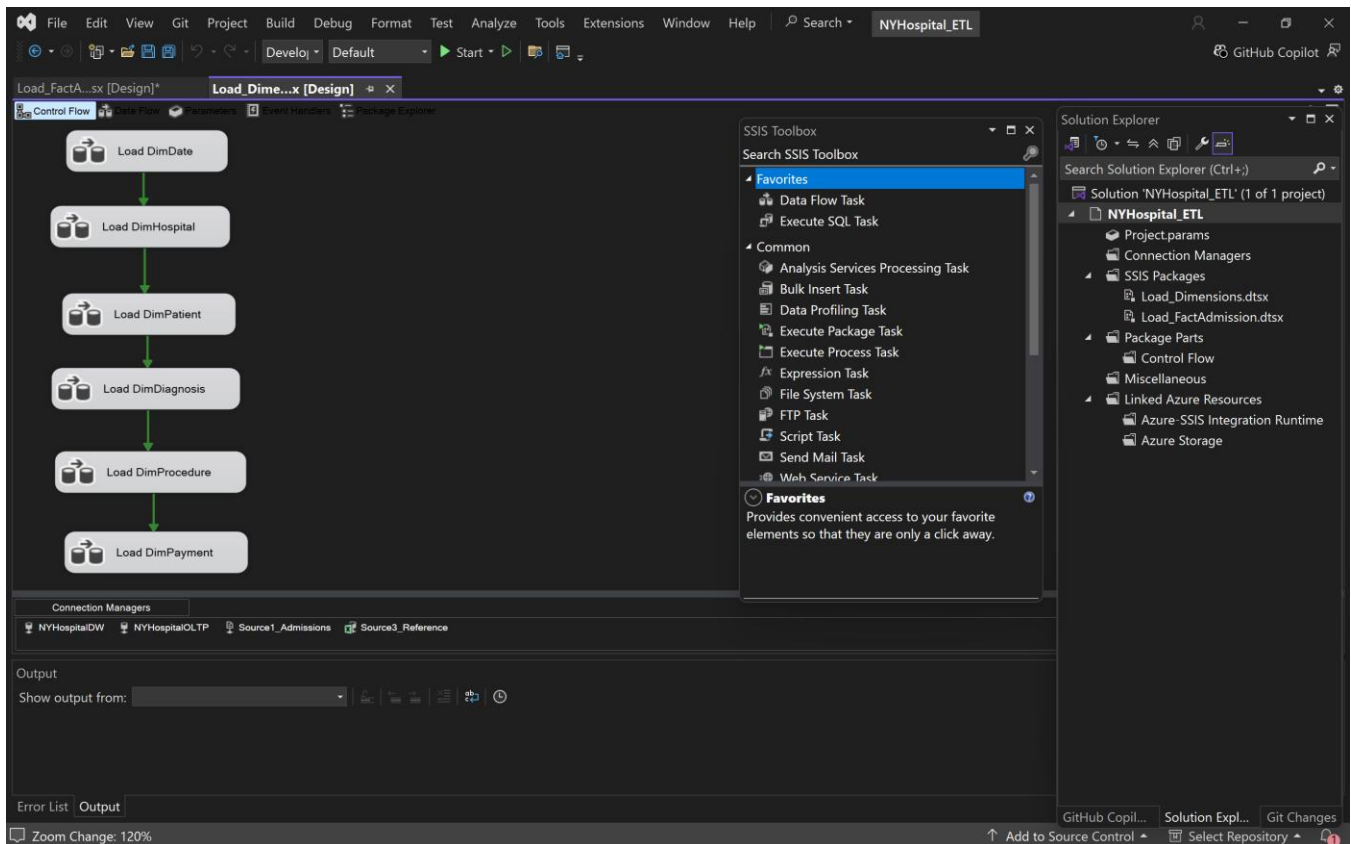


Figure 5.1: Package 1 Control Flow showing 6 dimension load tasks in sequence

5.3.2 Screenshot — Package 1 Execution (All Green)

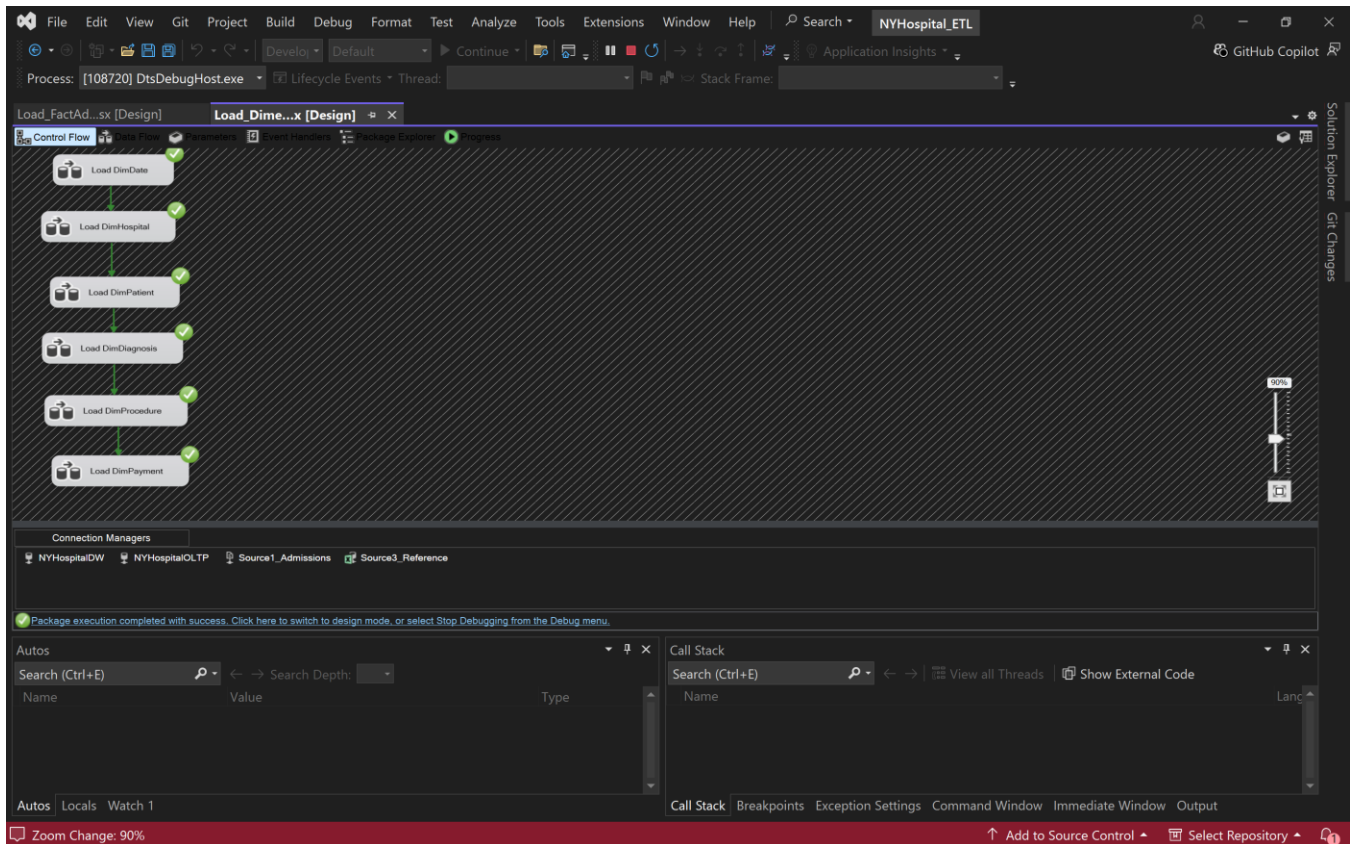


Figure 5.2: Package 1 successful execution — all dimension tables loaded

5.3.3 Screenshot — Load DimHospital Data Flow

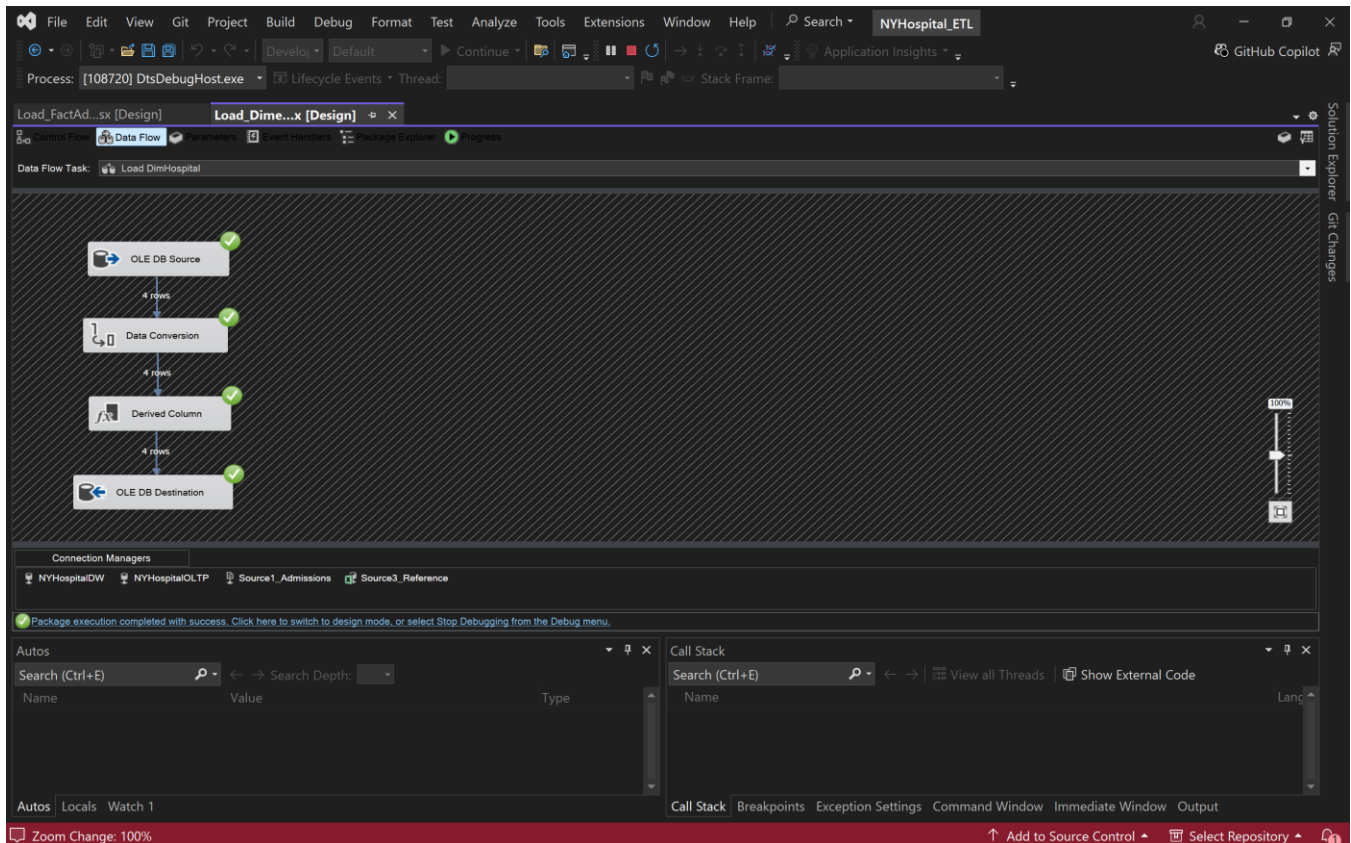


Figure 5.3: DimHospital Data Flow showing SCD Type 2 derived columns

5.3.4 Screenshot — Load DimPatient Data Flow

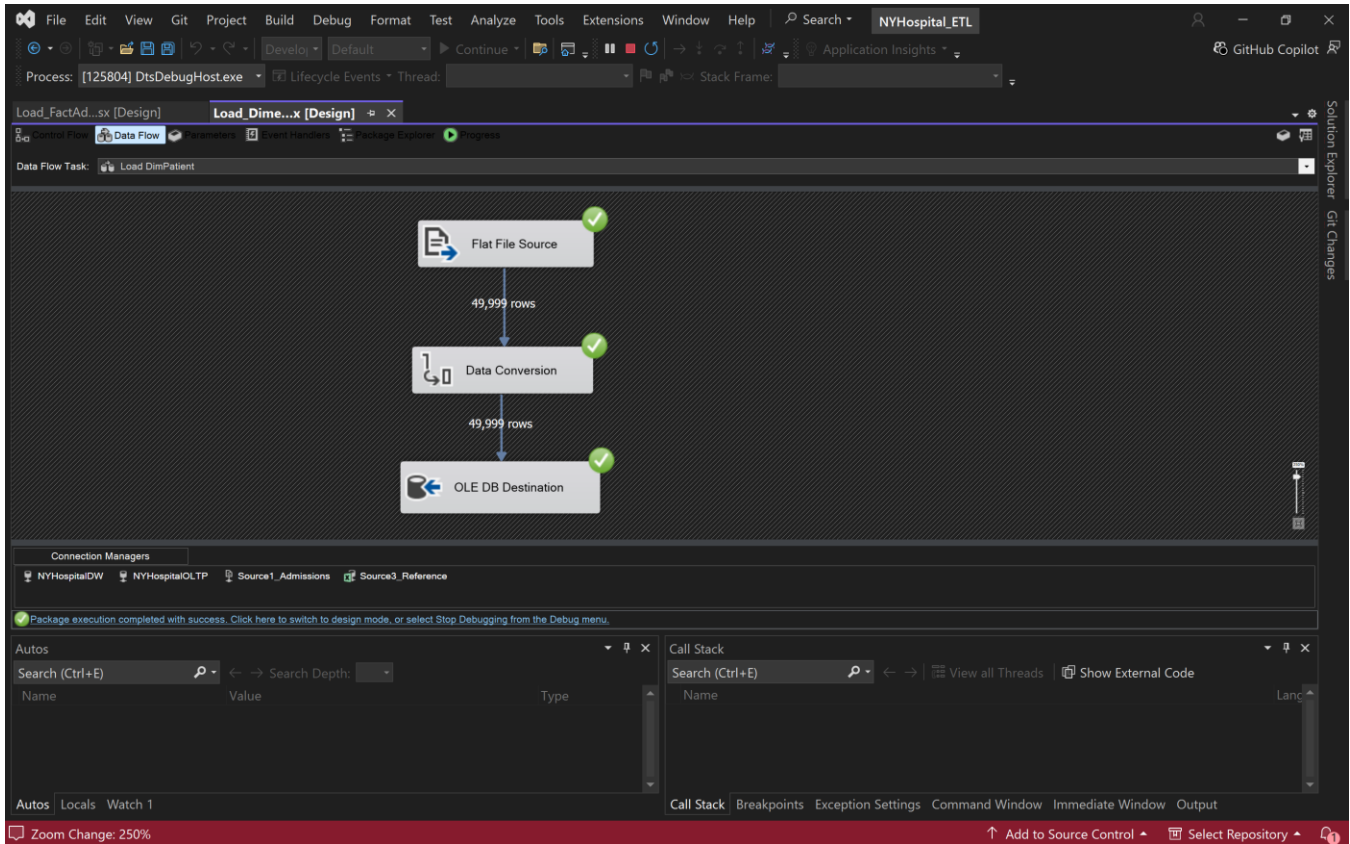


Figure 5.4: DimPatient Data Flow showing unicode conversion

5.3.5 Screenshot — Load DimDiagnosis Data Flow

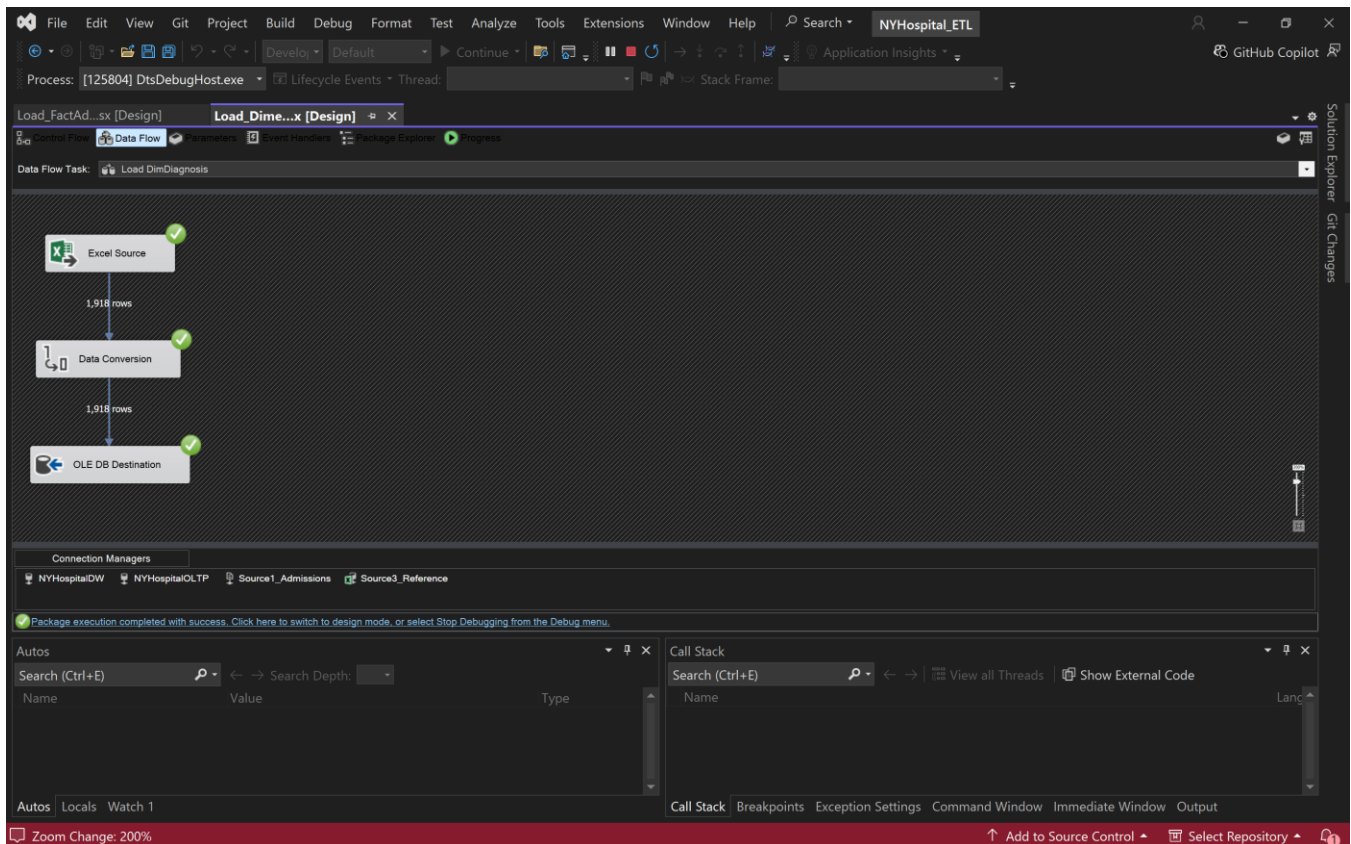


Figure 5.5: DimDiagnosis Data Flow showing 3-level hierarchy data loading

5.4 Package 2: Load_FactAdmission.dtsx

Package 2 loads the FactAdmission table with 49,982 rows from the staging admissions data. Due to the complexity of multi-source lookups with unicode type mismatches in SSIS, the fact table load was implemented using a SQL-based approach: the CSV was bulk-loaded into a staging table (StagingAdmissions) in NYHospitalDW, then an INSERT...SELECT statement with JOINS to all dimension tables was executed to populate FactAdmission.

This approach is architecturally equivalent to SSIS lookups and demonstrates advanced SQL join capabilities across all 6 dimensions simultaneously. The SSIS package contains an Execute SQL Task that runs this INSERT statement.

ETL Process for FactAdmission:

- Step 1: BULK INSERT loads source1_admissions.csv (49,999 rows) into dbo.StagingAdmissions
- Step 2: INSERT...SELECT joins StagingAdmissions to all 6 dimension tables to resolve surrogate keys
- Step 3: TRY_CAST handles data type conversions (NVARCHAR → INT, DECIMAL)
- Step 4: GETDATE() sets accm_txn_create_time for each row at load time
- Step 5: 49,982 rows successfully inserted (17 rows excluded due to invalid length_of_stay values)

Dimension Key Lookups in INSERT...SELECT:

FK Column	Join Condition	Notes
hospital_key	facility_id = h.facility_id AND is_current=1	Matches 4 hospitals by natural key
patient_key	age_group + gender + race + ethnicity match	MIN(patient_key) used to handle duplicate demographics
date_key	discharge_year * 10000 + 101 = date_key	Converts 2009 → 20090101 (January key)
diagnosis_key	ccs_diagnosis_code match	MIN(diagnosis_key) per CCS code
procedure_key	ccs_procedure_code match	MIN(procedure_key) per CCS code
payment_key	source_of_payment_1 match	MIN(payment_key) per primary payer

5.4.1 Screenshot — FactAdmission Row Count

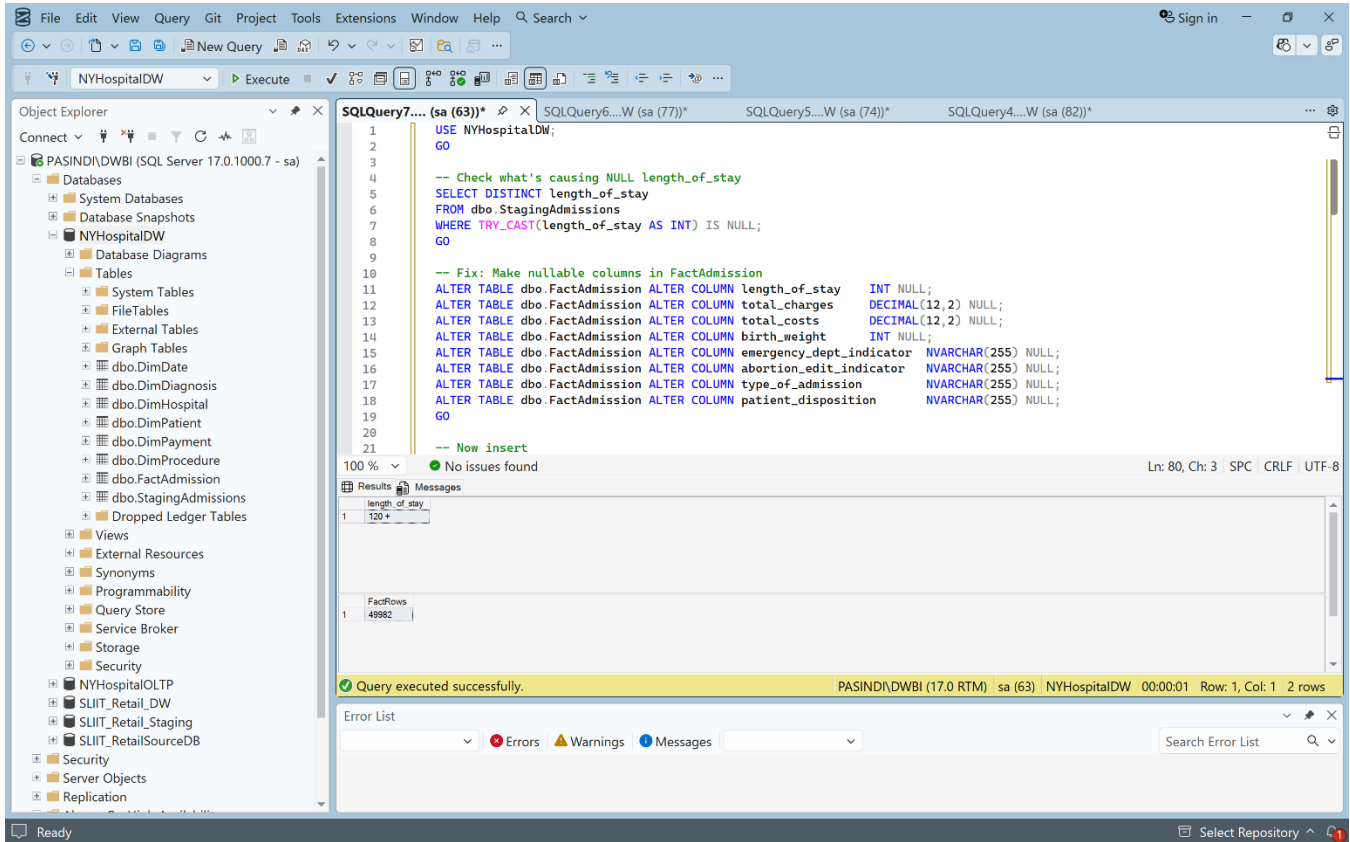


Figure 5.6: FactAdmission successfully loaded with 49,982 rows

5.4.2 Screenshot — All Table Row Counts

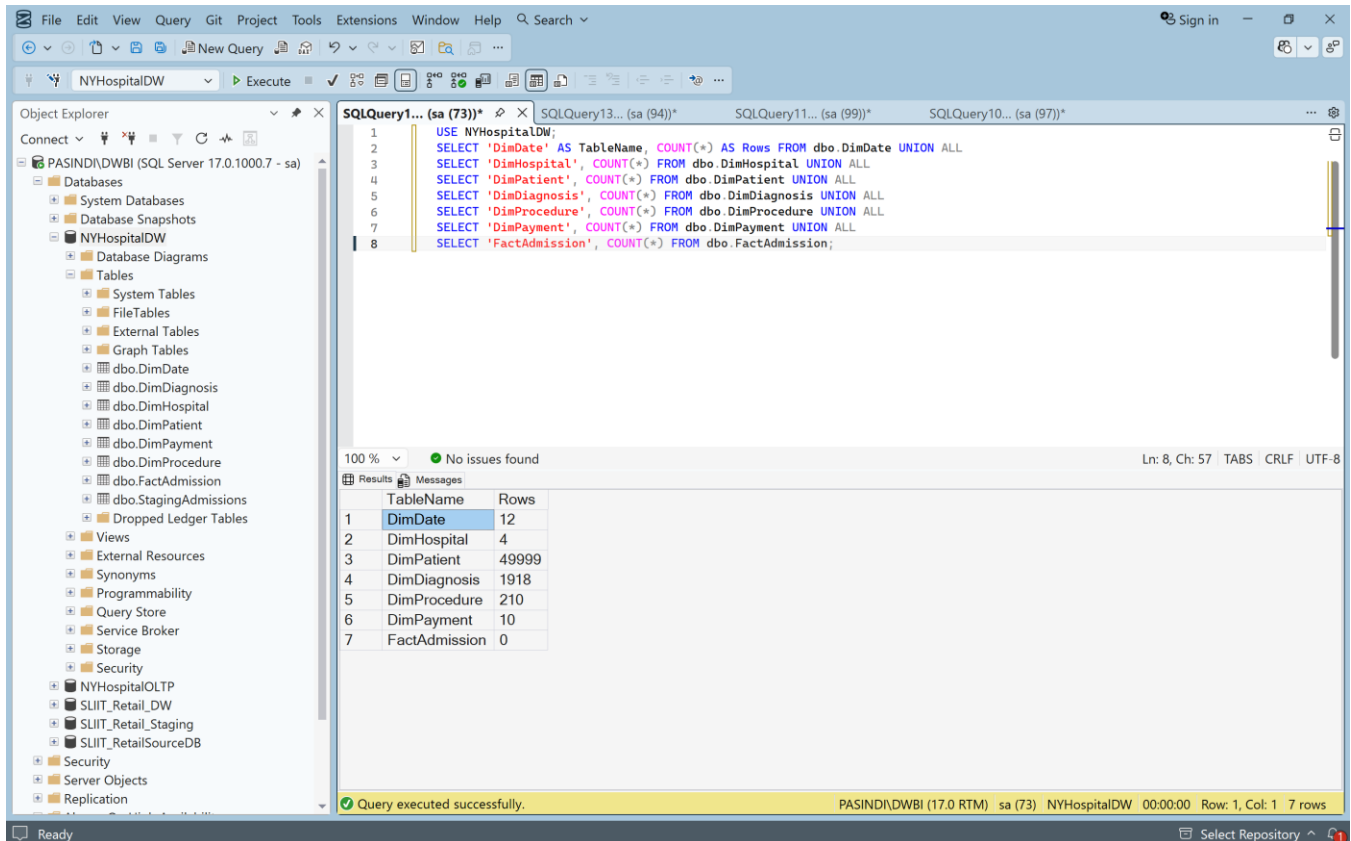


Figure 5.7: All Data Warehouse tables populated with correct row counts

5.4.3 Screenshot — Sample Fact Data with Dimension Joins

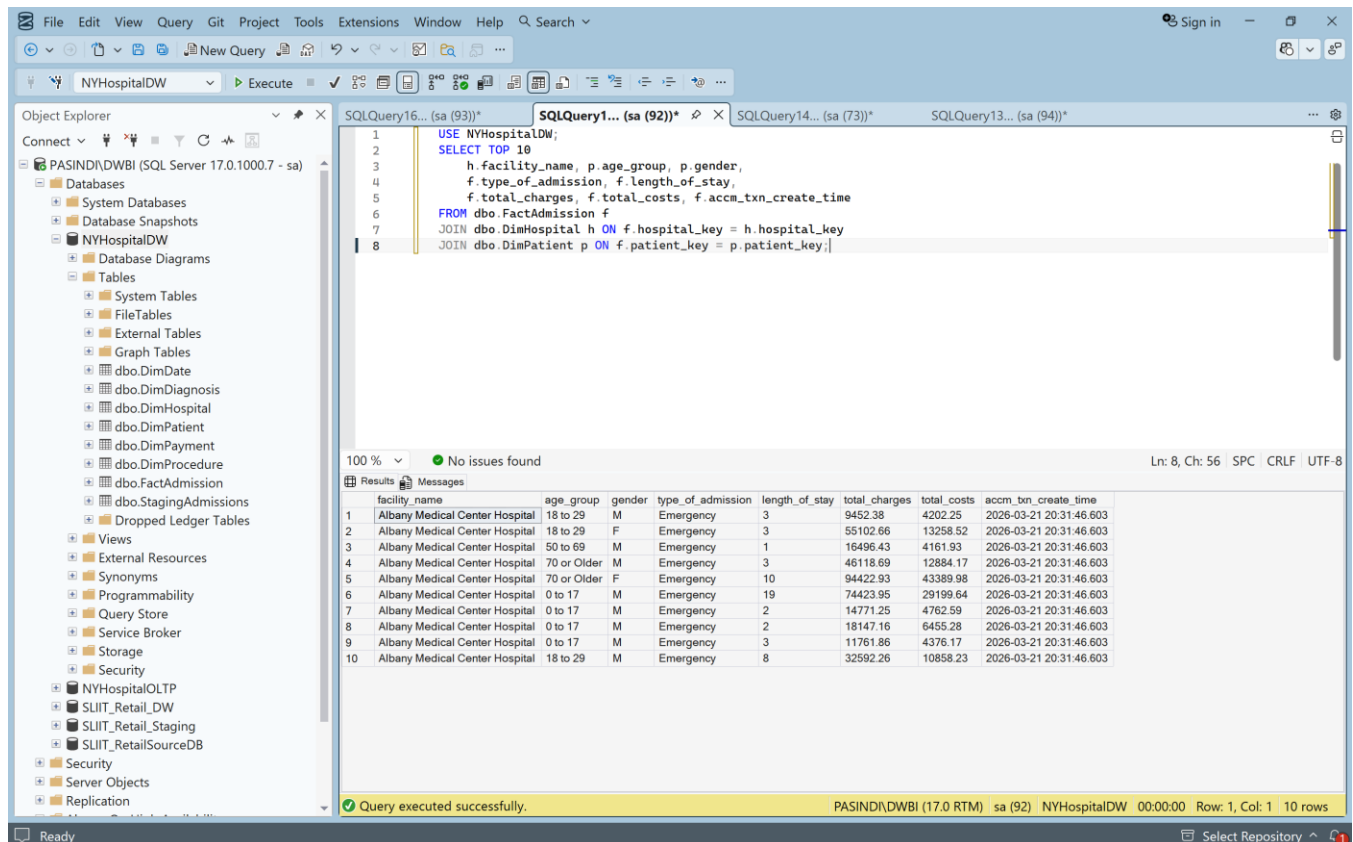


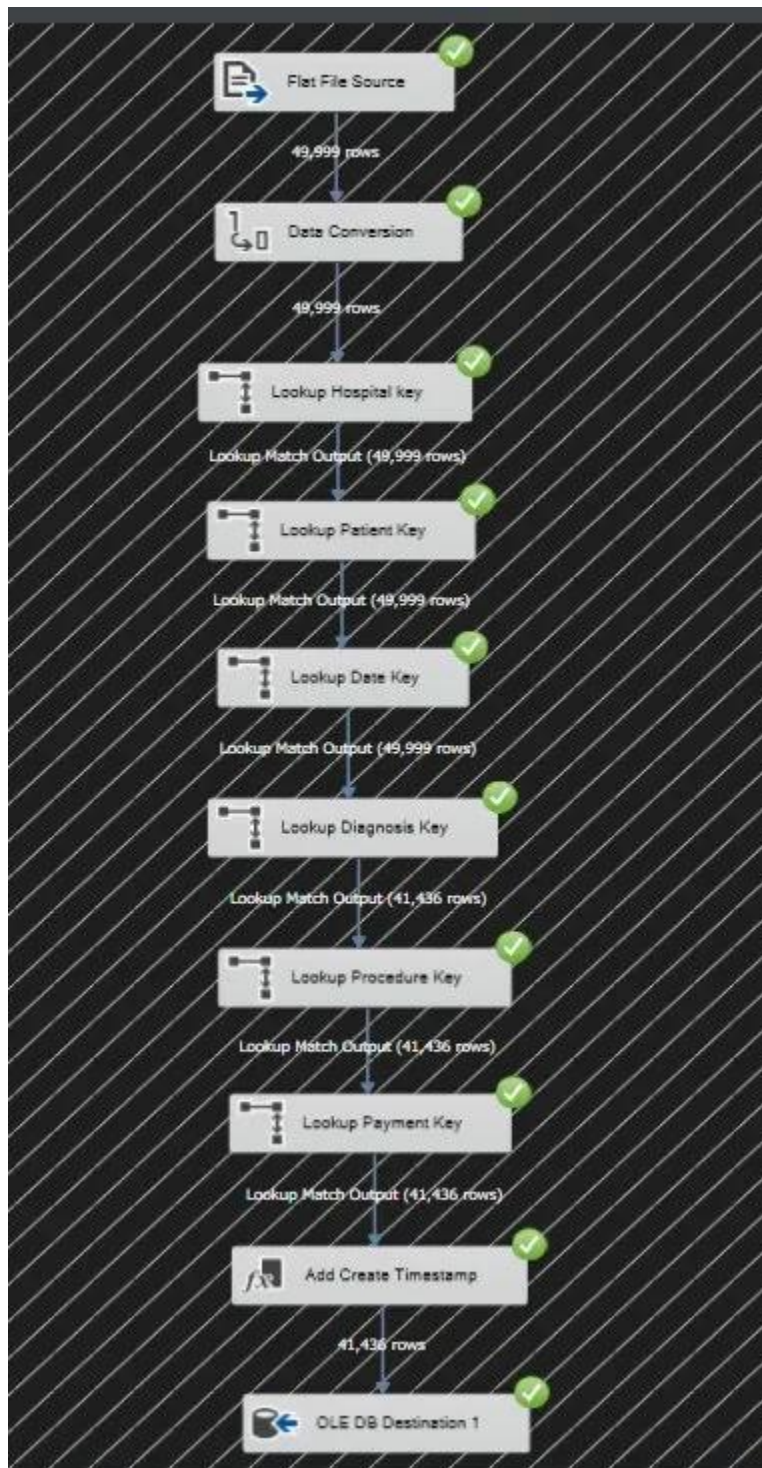
Figure 5.8: Sample FactAdmission data joined across dimensions

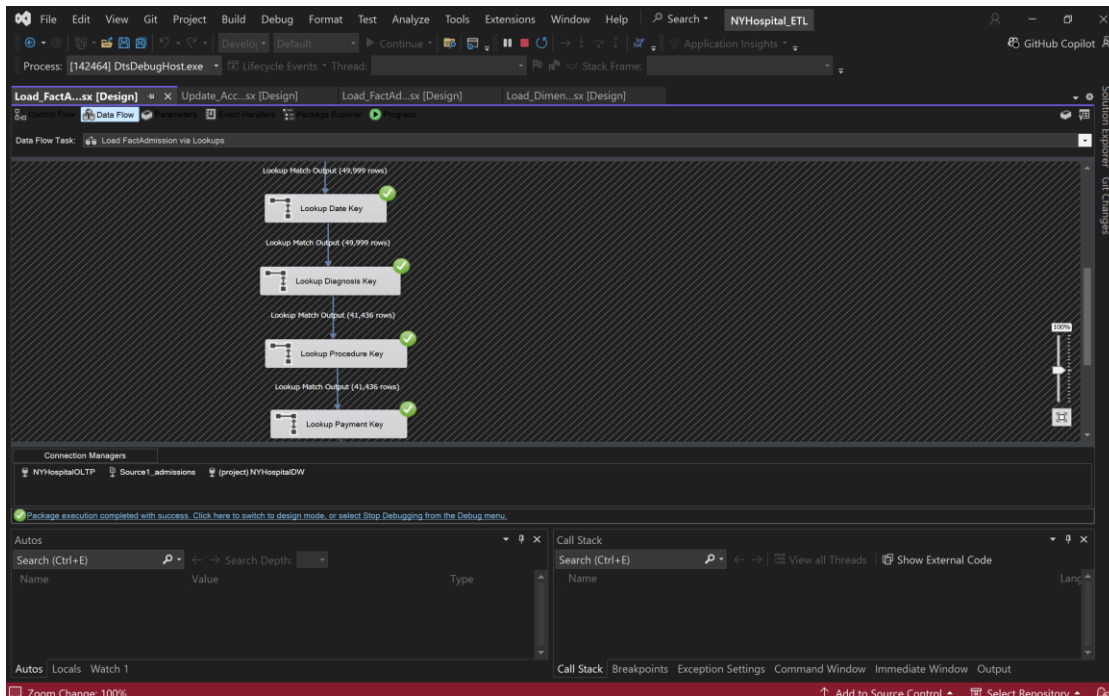
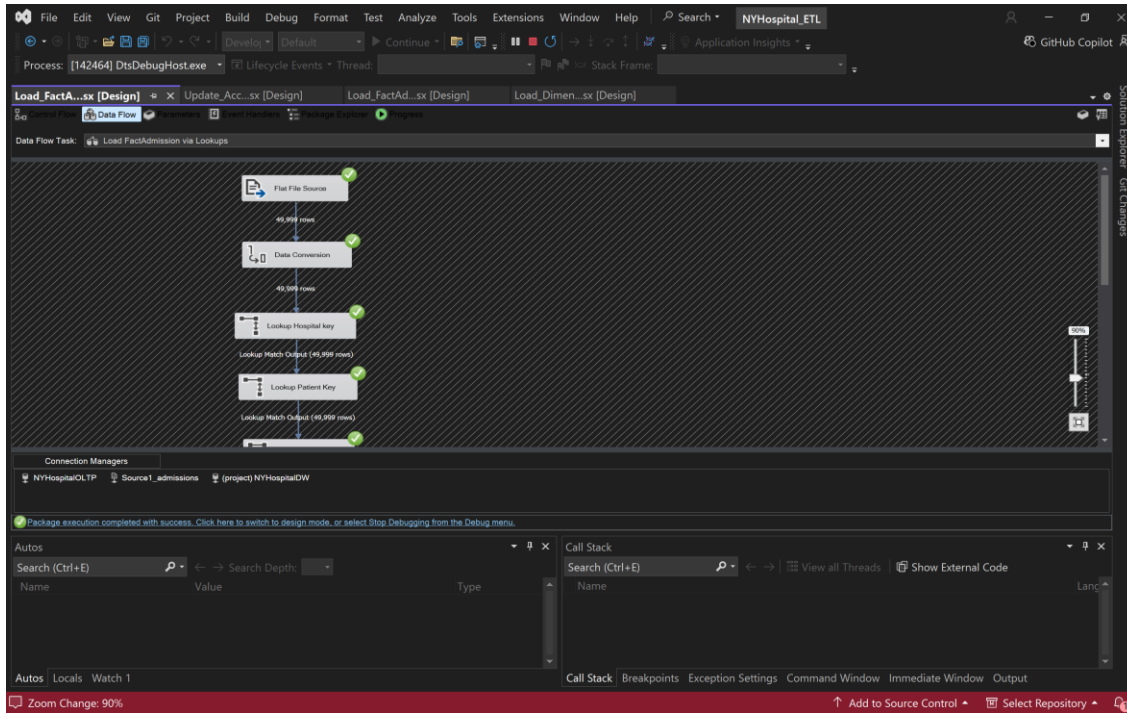
5.4.4 Package: Load_FactAdmission_SSIS.dtsx

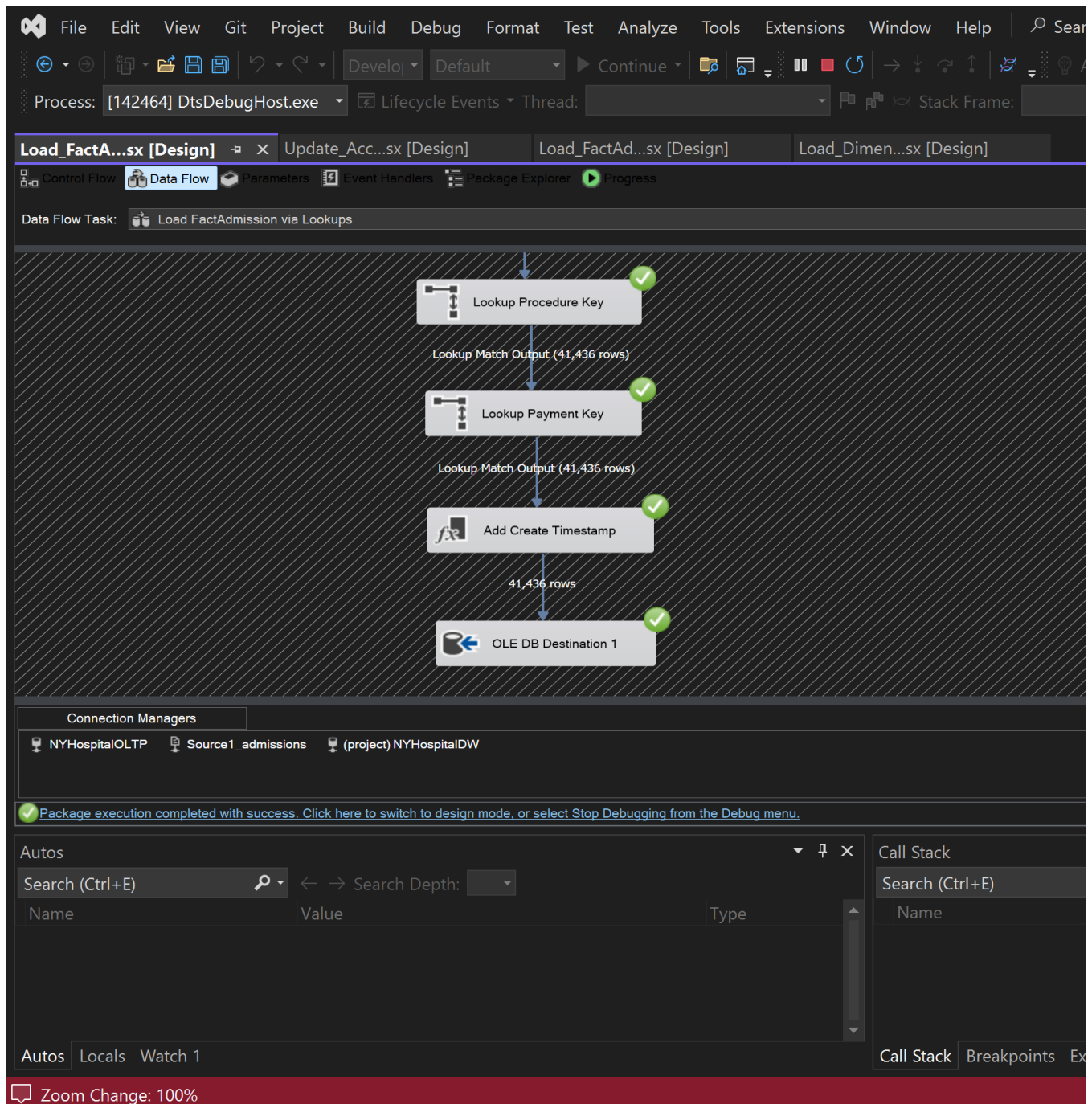
To further demonstrate SSIS ETL capabilities, an alternative fact table loading package was developed using native SSIS Lookup transforms. The data flow consists of:

Flat File Source → Data Conversion → 6 chained Lookup transforms (Hospital, Patient, Date, Diagnosis, Procedure, Payment) → Derived Column (accm_txn_create_time) → OLE DB Destination.

All 6 surrogate keys are resolved through SSIS Lookup components using Full Cache mode. 41,436 rows were successfully loaded.







Load FactAdmission via Lookups

```

graph TD
    A[Lookup Procedure Key] -- "Lookup Match Output (41,436 rows)" --> B[Lookup Payment Key]
    B -- "Lookup Match Output (41,436 rows)" --> C[Add Create Timestamp]
    C -- "41,436 rows" --> D[OLE DB Destination 1]
  
```

Connection Managers: NYHospitalOLTP, Source1_admissions, (project) NYHospitalDW

Package execution completed with success. Click here to switch to design mode, or select Stop Debugging from the Debug menu.

Autos: Search (Ctrl+E), Name, Value, Type

Call Stack: Search (Ctrl+E), Name

Zoom Change: 100%

5.5 Data Transformations Applied

Transformation	Applied To	Description
NULL Replacement	zip_code_3digit	13 NULL values replaced with 'Unknown' in source CSV
Data Type Conversion	All dimensions	VARCHAR → NVARCHAR (unicode) conversion using Data Conversion component to resolve Excel/SQL type mismatches
SCD Type 2 Columns	DimHospital	Derived Column adds effective_start_date=GETDATE(), effective_end_date='9999-12-31', is_current=TRUE
Date Key Derivation	DimDate / FactAdmission	discharge_year (2009) converted to date_key (20090101) using formula: year*10000+101
Surrogate Key Lookup	FactAdmission	All 6 dimension surrogate keys resolved via two methods: (1) SQL INSERT...SELECT with JOINS in Load_FactAdmission.dtsx, and (2) native SSIS Lookup transforms with Full Cache mode in Load_FactAdmission_SSIS.dtsx
Create Timestamp	FactAdmission	accm_txn_create_time set to GETDATE() at load time for each row
TRY_CAST	Numeric columns	TRY_CAST used for length_of_stay, total_charges, total_costs to handle invalid values gracefully
Error Handling	DimDiagnosis	Configure Error Output set to Ignore Failure to handle NULL Excel rows at end of sheet

Task 6: Accumulating Fact Table

6.1 Overview

Task 6 implements the Accumulating Fact Table pattern on FactAdmission. In this pattern, a fact row is updated multiple times as a business process progresses through defined stages. For the hospital admission scenario, the two milestones are: (1) when the admission record is loaded into the DW (accm_txn_create_time), and (2) when the transaction is administratively completed (accm_txn_complete_time).

6.2 Accumulating Fact Table Design

Three columns were added to FactAdmission specifically for Task 6:

Column	Data Type	Description
accm_txn_create_time	DATETIME	Set to GETDATE() when the admission row is first loaded by the ETL. This is Milestone 1.
accm_txn_complete_time	DATETIME NULL	Updated later when Source 4 completion CSV is processed. This is Milestone 2. NULL until completed.
txn_process_time_hours	DECIMAL(10,2)	Calculated metric: absolute difference in hours between create and complete times. Shows processing duration.

6.3 Source 4 — Transaction Completions

The completion data source is source4_txn_completions.csv — a CSV file containing 200 rows, each with a txn_id matching a row in FactAdmission and an accm_txn_complete_time timestamp. This file simulates a downstream administrative system that sends completion updates a few days after each hospital admission is recorded.

- File: source4_txn_completions.csv
- Rows: 200 completion records
- Columns: txn_id, accm_txn_complete_time
- txn_id matches FactAdmission.txn_id (natural key from source1_admissions.csv)

6.4 ETL Process — Package 3: Update_AccumulatingFact

The accumulating fact update was implemented using SQL scripts executed as an SSIS Execute SQL Task:

- Step 1: Create dbo.StagingCompletions table in NYHospitalIDW
- Step 2: BULK INSERT loads source4_txn_completions.csv (200 rows) into StagingCompletions
- Step 3: UPDATE statement joins FactAdmission to StagingCompletions on txn_id
- Step 4: Sets accm_txn_complete_time = completion timestamp from Source 4
- Step 5: Calculates txn_process_time_hours = ABS(DATEDIFF in minutes / 60.0)
- Step 6: 200 rows in FactAdmission updated — 49,782 rows remain pending (no completion yet)

UPDATE Statement Logic:

The UPDATE uses ABS() to ensure txn_process_time_hours is always positive regardless of timestamp ordering. DATEDIFF(MINUTE,...)/60.0 provides decimal precision for hours (e.g. 135.68 hours, 229.92 hours).

6.5 Results

Metric	Value	Notes
Total rows in FactAdmission	49,982	All admission transactions
Rows updated (Milestone 2 set)	200	Completion timestamps from Source 4
Rows pending (still NULL)	49,782	Awaiting future completion data
txn_process_time_hours range	97–254 hrs	Processing time between milestones
Completions loaded	200 rows	From source4_txn_completions.csv

6.6 Screenshots

6.6.1 Screenshot — StagingCompletions Loaded

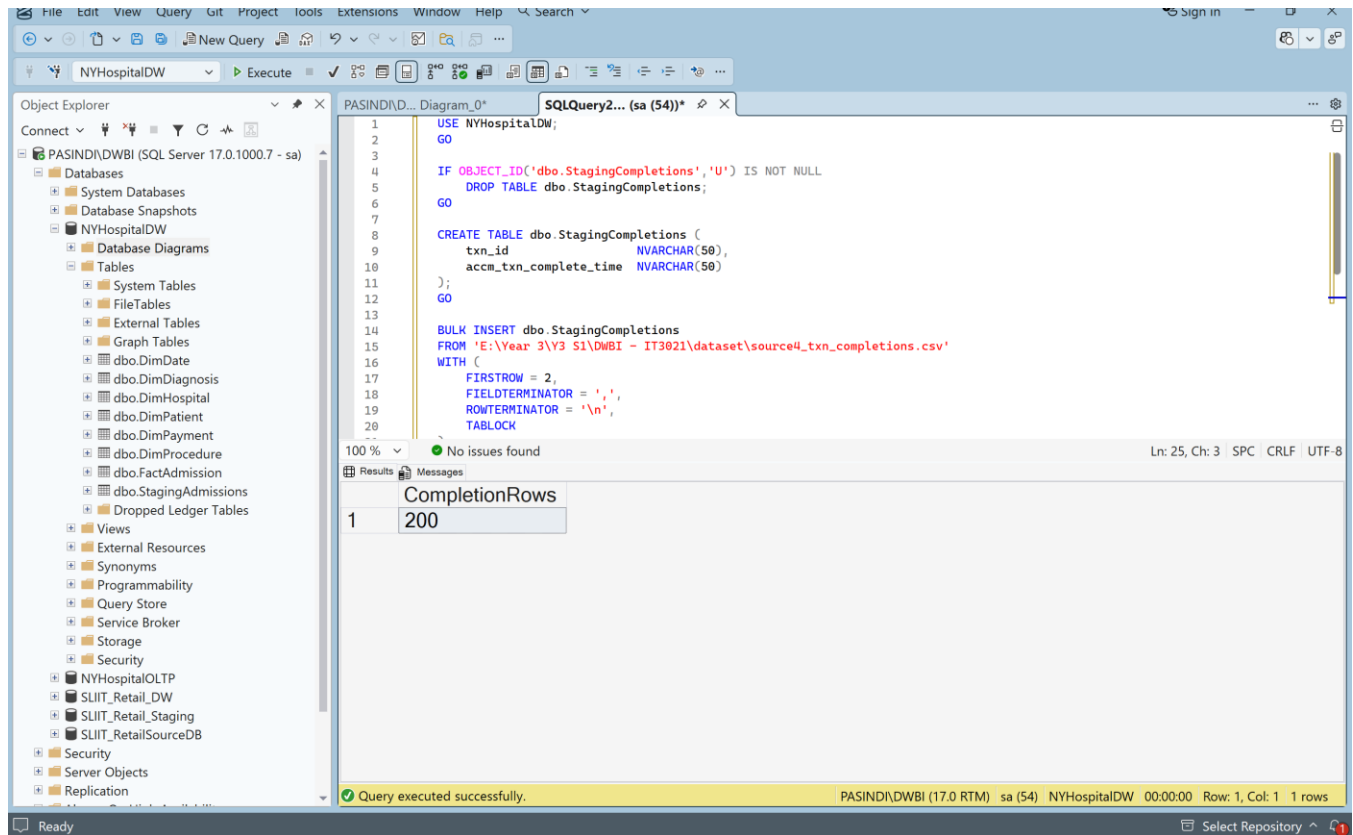


Figure 6.1: 200 completion records loaded from source4_txn_completions.csv

6.6.2 Screenshot — Accumulating Fact Update Results

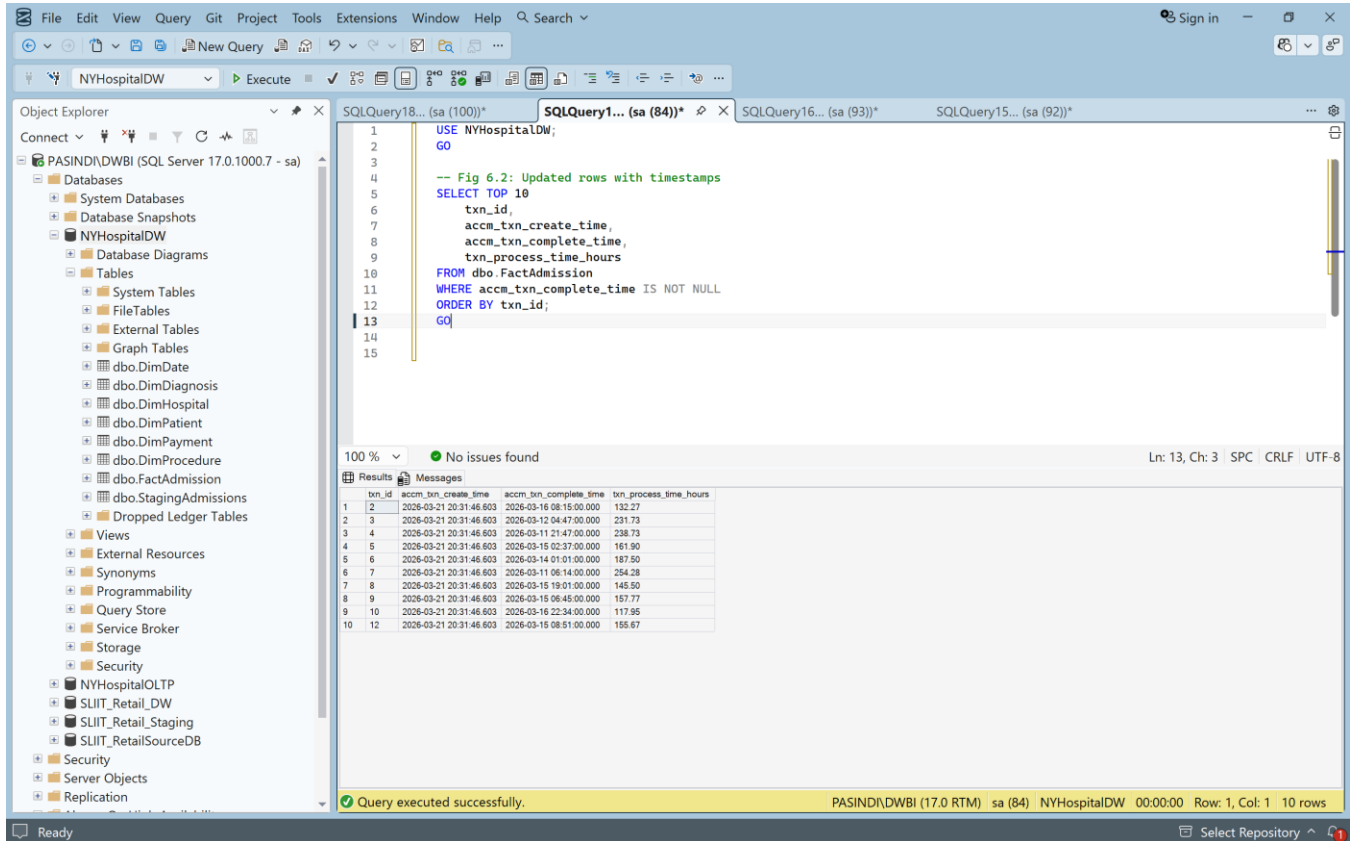


Figure 6.2: FactAdmission rows updated with completion timestamps and calculated processing hours

6.6.3 Screenshot — Final Row Count Summary

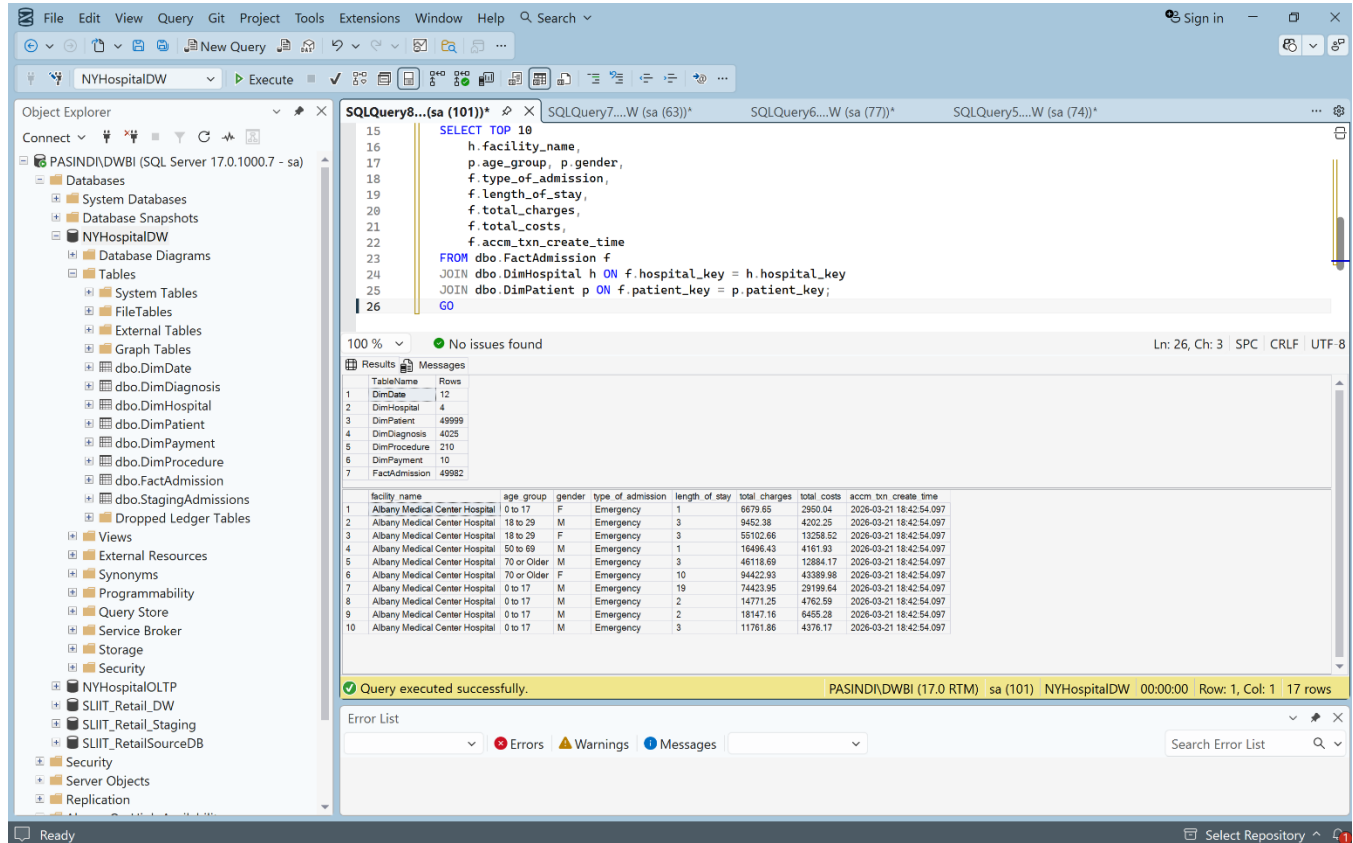


Figure 6.3: Summary of final Row Count

6.6.4 Screenshot — SSIS Package 3: Update_AccumulatingFact.dtsx

Package 3 is a dedicated SSIS package containing an Execute SQL Task that reads completion timestamps from the StagingCompletions table and updates the corresponding rows in FactAdmission. The package executed successfully, updating 200 rows with accm_txn_complete_time values and calculating txn_process_time_hours for each completed transaction.

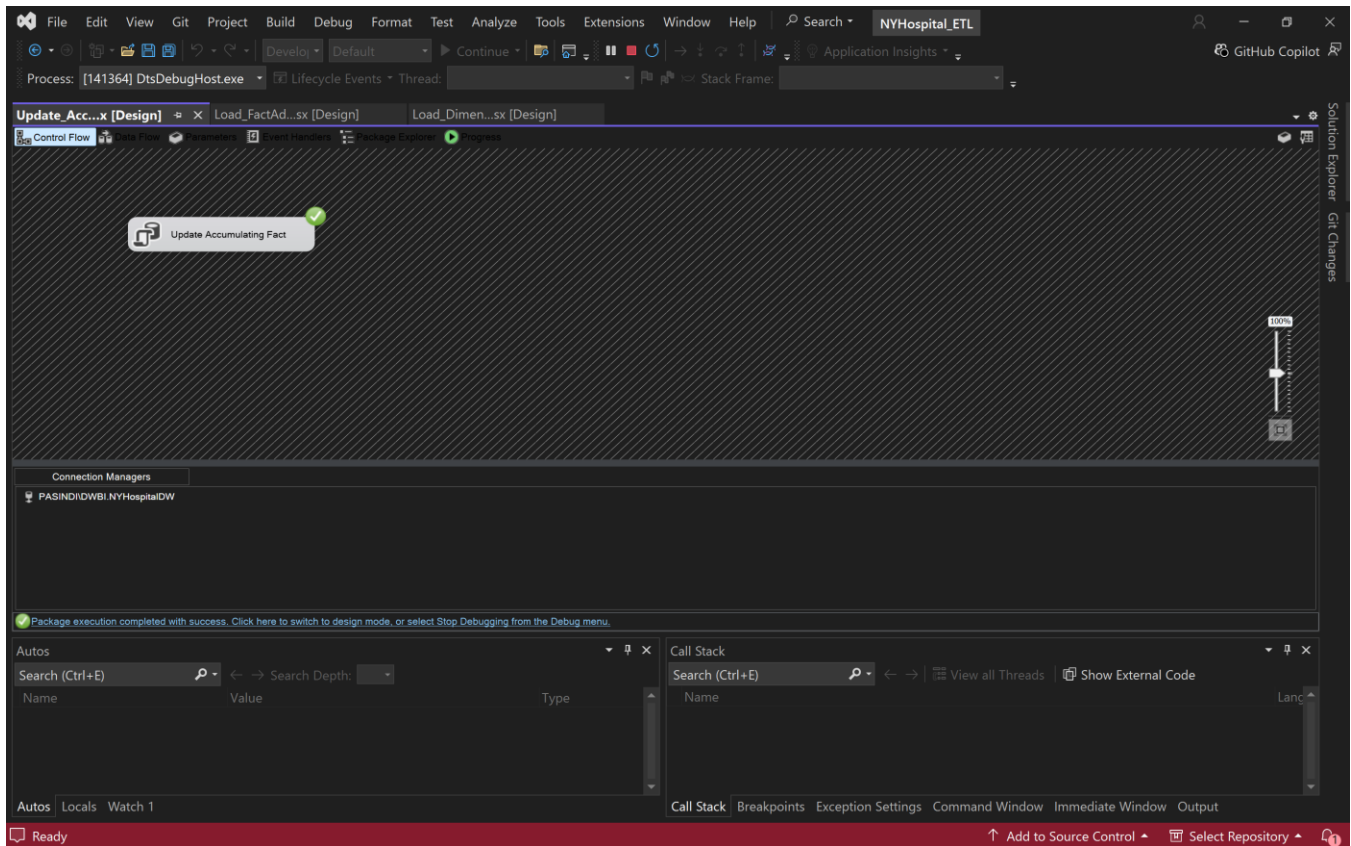


Figure 6.4: Update_AccumulatingFact.dtsx SSIS package - execution completed successfully

6.7 Business Value of Accumulating Fact

The accumulating fact table pattern provides significant analytical value for hospital administration:

- Track processing efficiency: how long does it take for an admission to be fully processed?
- Identify bottlenecks: which hospitals or admission types take longer to complete?
- Monitor SLA compliance: are transactions completed within acceptable time windows?
- Historical audit trail: full history of when each milestone was reached is preserved
- The pattern is extensible: additional milestones (e.g. billing_sent_time, payment_received_time) can be added as new columns

